

In Defense of the Clause-Internal Phase: Evidence from Operational Opacity in Ndebele

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Abstract

The partition of clause structure into two locality domains (CP and vP) has been standard in syntactic theory since the conception of Phase Theory (Chomsky 2000, 2001). A common type of evidence for the domains comes from footprints of successive cyclic movement, observed at the edges of both CP and vP. However, the two domains famously differ in their opacity for syntactic operations such as A-movement and ϕ -agreement, with vP being significantly more transparent than CP. This asymmetry has led some authors to question the phasehood of vP, especially in the context of alternative explanations for intermediate movement through vP (i.e. Keine & Zeijlstra to appear). This paper argues that we need not rely on intermediate movement as evidence for a clause internal phase – operational opacity is likewise observed at the clausal middle field. Evidence presented here comes from Zimbabwean Ndebele, in which VoiceP is argued to be opaque for A-movement and ϕ -agreement. The paper considers two alternative explanations for the absence of A-relations across VoiceP: one in terms of the failure of probing and the other as the absence of a matching goal. Both are ruled out on empirical grounds.

1 Introduction

Phase Theory (Chomsky 2000, 2001) posits that each clause consists of two locality domains (phases): the complete clause (CP) and a clause internal domain located roughly around the vP. While there appears to be no controversy regarding the phasehood of the CP (though see den Dikken 2009, 2017), the clause internal phase remains a subject of debate. This paper contributes new evidence from Zimbabwean Ndebele in support of the clause-internal phase hypothesis, arguing that VoiceP in this language is opaque for A-movement and ϕ -agreement.

Syntactic evidence for the phasehood of CPs has come primarily from two types of phenomena: operational opacity and footprints of successive cyclic movement. Operational opacity is typically observed in A-relations, such as A-movement and ϕ -agreement. A well-known example is A-movement in English, which can cross a TP (11) but not a CP (12).

- (1) a. **Abby** seems [_{TP} <Abby> to be home].
b. ***Abby** seems [_{CP} (that) <Abby> is home].

The phasehood of CP is corroborated by the existence of successive cyclic effects at the edge of CP. The example below comes from Belfast English, in which CPs targeted by wh-movement exhibit subject-auxiliary inversion (2-a). Inversion is also found in clauses which do not have an overt wh-phrase in Spec,CP, as is the case in the embedded CP in

(2-b). Crucially, the embedded CP in this sentence has been crossed by wh-movement. This fact has been interpreted as evidence that long-distance wh-movement has a step in the embedded Spec,CP, resulting in subject-auxiliary inversion in both the embedded and the matrix clause.

- (2) a. I asked them [_{CP} **what**_i had they done t_i]?
 b. **What**_i did Mary claim [_{CP} t_i did they steal t_i]? (Henry 1995)

It is important to note that the occurrence of subject-auxiliary inversion does not entail a successive cyclic derivation of long distance wh-questions. Successive cyclic movement is an analysis of this effect (see, for instance, Adger & Ramchand (2005), who derive successive cyclic complementizer effects in Scottish Gaelic without positing intermediate movement; see also den Dikken 2017 for a critical review of crosslinguistic evidence of this type). The general consensus around the phasal status of CPs is largely due to the concurrent evidence from operational opacity, i.e. the fact that a CP boundary blocks many syntactic processes.

Syntactic evidence for a clause internal phase abounds as well, and it comes primarily from successive cyclic effects (i.a. Barbiers 2002; Legate 2003; Sauerland 2003; Aldridge 2008b; Bennett et al. 2012; Henry 2012; van Urk 2015; Erlewine & Lim 2023). The example from Dinka in (3) illustrates a footprint of wh-movement at the edge of every crossed vP. In van Urk & Richards 2015, the pronoun *ké* has been analyzed a copy of the moved wh-phrase, revealing intermediate steps of this movement through vP.

- (3) *Dinka Bor: A wh-copy left in every vP crossed by movement*
 [Yeyínà] yé [_{vP} **ké** tâak, [_{CP} cí Bòl [_{vP} **kè** tîŋ]]?
 who._{PL} HAB.2SG _{PL} think PFV.NSV Bol.GEN _{PL} see
 ‘Who all do you think Bol saw?’ (van Urk & Richards 2015:128)

Despite its abundance, evidence from intermediate movement has been called into question by Keine & Zeijlstra (to appear), who point out that intermediate movement does not unambiguously diagnose phases. The phasehood of some category X is just one possible reason why movement passes through the specifier of XP. Another reason, suggested by Keine & Zeijlstra, is the need to obviate a minimality violation. An important fact about Dinka, as well as some other languages from which similar evidence has been presented, is that $\bar{A}$ -movement is restricted to target the highest argument. If a lower argument is to be extracted, it must first move across the highest argument to avoid intervention. This is schematized in (4).

- (4) 

The first step of movement in (4), often referred to as leapfrogging (McGinnis 1998), is an intermediate step of movement required for reasons having nothing to do with phases.

Keine & Zeijlstra argue that successive cyclic movement in Dinka, as well as a number of other languages, is precisely the intervention-obviating movement shown in (4). Given the existence of a reasonable alternative to the phase-based analysis, they conclude that

intermediate movement does not constitute strong evidence for the phasehood of vP.

As with CP, a stronger case for phasal vP could be made if we could demonstrate clear operational opacity of this category. In this respect, however, CP and vP differ – we observe significantly more opacity effects at the CP boundary than at the vP boundary. A well-known example is A-movement, for which CPs tend to be opaque crosslinguistically, but vP is notoriously transparent. A similar generalization can be made for ϕ -agreement. The apparent absence of operational opacity at the clausal middle field has first led to the weakening of the Phase Impenetrability Condition so that vP-internal material is accessible from T (Chomsky 2001). Eventually, this and other types of evidence led to the rejection of the vP phase hypothesis altogether (Grano & Lasnik 2018, Keine 2020a,b, Mendes & Ranero 2021, Keine & Zeijlstra to appear).

This article argues for the clause internal phase hypothesis based on the crucial type of evidence from operational opacity. Specifically, it is argued that VoiceP in Zimbabwean Ndebele is opaque for A-movement and ϕ -agreement. In section 2, I present the core data and the analysis in terms of phasal VoiceP. In brief, it is argued that the absence of subject movement to Spec,TP is due to the inability of T to locate the subject across an intervening VoiceP. In the same section, I discuss two correct predictions that follow from this analysis. I then consider two other possible explanations for the absence of subject movement: i) the lack of probing by T (section 3) and ii) the lack of a suitable goal (section 4). I argue that both types of analysis make incorrect predictions and that the phasal VoiceP account derives the full ranges of facts. In section 5, I offer some remarks on how the VoiceP phase proposal relates to broader debates about the locus and existence of a clause-internal phase, as well as the nature of successive cyclic movement. Section 6 concludes.

2 Proposal: Phasal VoiceP blocks A-movement and ϕ -agreement

Subjects in Ndebele move to Spec,TP and control agreement on T, as shown in (5).¹

- (5) UZondi_i u-a-phek-a [_{vP} t_i t_V inyama].
 1Zondi 1s-PST-cook-FV 9meat
 ‘Zondi cooked meat.’

However, this movement need not take place. All subjects, even in transitive clauses, can remain in-situ, giving rise to so called (Transitive) Expletive Constructions, found in many other Bantu languages. As seen in (6), an in-situ transitive subject follows the verb (due to head movement of the verb out of vP) and precedes the object. Note that in-situ subjects must be interpreted as discourse-new, as indicated by the context provided in (6).

¹ Unless otherwise notes, data presented in this paper were collected by the author through elicitation of acceptability judgments. All data comes from a single speaker to ensure that it represents a coherent mental grammar. The consultant is a native speaker of Ndebele (age 60) who grew up in Bulawayo, Zimbabwe, and currently resides in the US. The Leipzig convention is used for glosses with the following additions: 1 – class 1 nominal prefix (etc.), 1s – class 1 subject agreement (etc.), 1o – class 1 object agreement (etc.), A – augment vowel, CNJ – conjunct, DSJ – disjoint, FV – final vowel.

- (6) Ku-a-phek-a [_{VP} uZondi t_V inyama].
 15s-PST-cook-FV 1Zondi 9meat
 ‘Zondi cooked meat.’ (Answers: Who cooked meat?/What happened?)

The SV-VS alternation is very productive and is not sensitive to the type of predicate or the thematic properties of the subject. For instance, unaccusative subjects exhibit the same optionality of movement to Spec,TP (7) (see Pietraszko 2023b for a full range of facts).

- (7) a. I-sihlahla s-a-w-a.
 A-7tree 7s-PST-fall-FV
 ‘The tree fell.’
 b. Kw-a-w-a i-sihlahla.
 15s-PST-fall-FV A-7tree
 ‘A tree fell.’ (Pietraszko 2023b:8-9)

As mentioned above, in-situ subjects in Ndebele cannot be interpreted as topical. Instead, they may either form part of a broad focus interpretation or receive narrow focus (as indicated in (6)). While it might be tempting to link A-movement in Ndebele to a discourse-related feature on the moving DP, such as Antifocus (Zeller 2008, 2015), prior work has shown that A-movement in this language applies to both focused and non-focused DPs (Pietraszko 2021; see also section 4). The discourse properties of preverbal subjects in Ndebele are analyzed in Pietraszko 2017a, 2021, where it is proposed that subjects in indicative clauses move to Spec,TopP, accounting for the observation that only indicative clauses require their preverbal subjects to be topical. What remains to be explained is why in-situ subjects must be discourse-new. While I don’t commit to a concrete answer here, it is worth noting that this restriction is attested crosslinguistically, and so its explanation is likely to be more general in nature, applying across languages, and therefore not tied to the specific mechanics of A-movement in Ndebele.²

Returning to the optionality of subject movement, the fact that the subject can some-

² Analyzing Xhosa, Carstens & Mletshe 2015 observe several restrictions on VSO constructions like (6), which they refer to as Transitive Expletive Constructions (TECs). Specifically, they report that i) subjects in TECs must be narrowly focused, ii) independent pronominal objects are disallowed, and iii) experiencer verbs do not appear in TECs. They propose an analysis of in-situ subjects that derives these properties. In Ndebele, however, the first two restrictions do not hold: subjects in TECs need not be narrowly focused and may be part of a broad focus interpretation, as indicated in (6). Further, independent pronouns are freely allowed as objects:

- (i) Kupheke uZondi yona.
 15-cooked 1Zondi 9PRON
 ‘Zondi cooked it.’ [Answers e.g.: Was the meat(cl.9) cooked?]

The third restriction – that experiencer verbs do not occur in TECs – is partially reflected in Ndebele: some experiencer verbs require a preverbal subject, as in Xhosa, while others do not:

- | | |
|---|---|
| (ii) Kw-azi uZondi isiKhiwa.
15s-know 1Zodni 7English
‘Zondi knows English’ | (iii) *Ku-funa uZondi ikhekhe.
15s-know 1Zodni 9cake
‘Zondi wants cake’ |
|---|---|

times move to Spec,TP indicates that T and the subject can enter in an A-relation that results in movement. This raises the question of why this relation does not hold in (6) and (7-b). Syntactic theory offers three types explanations for the absence of a relation:

(8) Possible sources of non-relation

- i. Lack of probing
- ii. Lack of a suitable goal
- iii. A locality issue

The first possible reason why a probe-goal relation is not established is simply the absence of a probe. In this vein, we could derive VSO sentences by positing that T lacks EPP in those contexts. Conversely, we might assume that T always has the movement feature but the subject is not always a matching goal for it. I will discuss both analyses in sections 3 and 4 respectively and will demonstrate that neither meets empirical adequacy. I will argue that the third option, which appeals to syntactic locality, derives the core facts in an equally straightforward way, but makes correct predictions where the alternatives fail. As detailed below, the locality issue is a clause-internal phase.

2.1 Analysis

A phase-based analysis of VS orders in Ndebele was first proposed by Pietraszko (2023b), who argues that in-situ subjects are inaccessible to T due to an intervening phasal VoiceP. This category is argued to dominate all arguments in their base positions (see Pietraszko 2023b for a detailed discussion and argumentation;³ see also Zeller (2014) for a proposal that a phasal XP dominates the entire argument structure domain in closely related Zulu.) Following existing literature on related Bantu languages, Pietraszko assumes that T in Ndebele always has EPP (i.a. Carstens 2005; Halpert 2015; Zeller 2015). The key to explaining the optionality of subject movement is the idea that successive cyclic movement through the edge of phasal VoiceP is optional. When it takes place, it feeds further movement to Spec,TP, as shown in (9). When it does not take place, the subject is trapped inside the VoiceP phase and is thus invisible to T (10).⁴ The optionality of movement to Spec,VoiceP is implemented as an optional EPP feature in Voice.⁵ Note that, in order to derive a VS

I do not currently have an explanation for this variation, but it may indicate that certain verbs idiosyncratically require a VoiceP with an EPP feature. I must leave the contrast above to future research, though I note that Carstens and Mletshe's analysis would likely not be applicable here, given the broader empirical differences between TECs in the two languages.

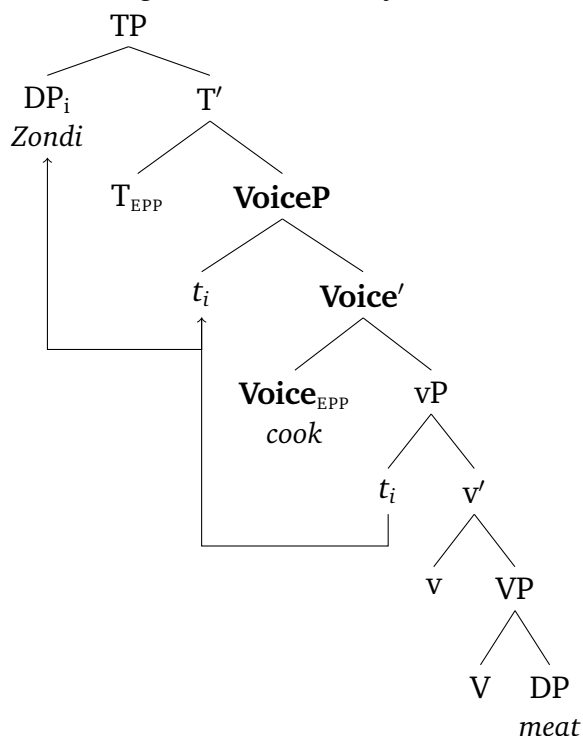
³ VoiceP is taken here to be the category that encodes the active-passive alternation, not as the head that introduced the external argument (Collins 2018). External arguments are introduced either by *v* or by Cause. It is important to note that the arguments presented in this paper are not on contingent on this precise choice of label for the clause internal phase, only on the assumption that the phase head embeds the entire argument structure domain.

⁴ The analysis assumes the original, stronger version of the Phase Impenetrability Condition, whereby a phasal complement is inaccessible to the next phase up (Chomsky 2000).

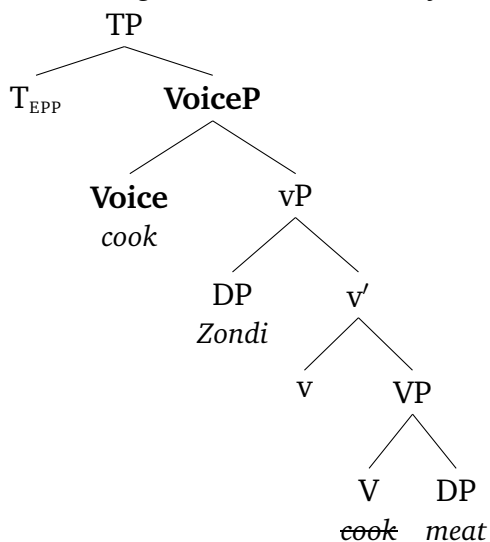
⁵ Pietraszko argues that Voice additionally has a ϕ -probe, which agrees with the object, if one is present, or it doesn't agree at all. Object agreement is immaterial in the present discussion and I will not indicate the ϕ -probe on Voice in the derivations. I refer the reader to Pietraszko 2023b for a full account of object agreement in the context of the phasal Voice hypothesis.

order when the subject stays in situ, the verb must move to a position higher than the base position of the subject. As we will see in section 2.3, this verb movement takes place even in the presence of auxiliary verbs, suggesting that it targets a low position, such as Voice.

- (9) Voice merged with EPP: subject visible to T from Spec, VoiceP



- (10) Voice merged without EPP: subject stays in vP and is invisible to T



Note that, when the subject stays in situ, the EPP feature in T remains unchecked without causing a violation. This can be understood in one of two ways. We may adopt a framework in which unchecked features do not cause a derivational crash (Preminger 2014). Alternatively, one might posit that a null expletive is merged in Spec,TP just in case the EPP feature fails to locate a goal in the structure. As I argue in section 3.2, the idea of an

expletive *pro* is problematic for Ndebele. I therefore assume EPP may remain unchecked.

2.2 Prediction 1: Subject movement and agreement go hand-in-hand.

This analysis correctly predicts a correlation between subject agreement and subject movement to Spec,TP. In-situ subjects in Ndebele cannot control agreement on T. Instead, the agreement prefix on the verb must be the default class 15 prefix (11).

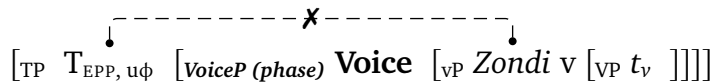
- (11) **Ku/*U-Ø-phek-a** [_{VP} uZondi *t_V*].
15s/*1s-CNJ-cook-FV 1Zondi
 ‘Zondi cooks.’

In contrast, subjects that move to Spec,TP control agreement on T obligatorily – class 15 agreement is not allowed (12).

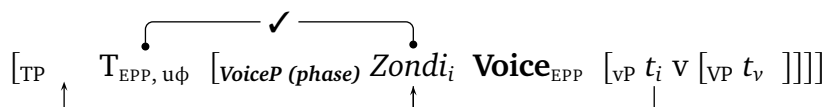
- (12) UZondi_i **u/*ku-ya-phek-a** [_{VP} *t_i* *t_V*].
 1Zondi **1s/*15s-DSJ-cook-FV**
 ‘Zondi cooks.’

This bi-conditional relation between subject agreement and movement to Spec,TP follows directly from this account. Assuming that both operations are triggered by the features of T, the subject is accessible to either both of them or to neither of them. When the subject stays in-situ, it is inaccessible to both EPP and ϕ in T (13). When it moves to the edge of phasal VoiceP, it becomes exposed to both probes in T (14).

- (13) *In-situ subject is inaccessible to the probes in T due to phasal VoiceP*



- (14) *Subject in Spec, VoiceP is accessible to the probes in T due to being at phase edge*



Note that this explanation assumes that operations are obligatory when possible. Since the subject in Spec, VoiceP is visible to T, further movement to Spec,TP and agreement with T are derivationally inevitable.

Previous accounts have invoked special mechanisms to explain the correlation between agreement and movement in Bantu languages. For instance, Carstens (2005) implements this correlation by positing that ϕ -probes have an EPP property/subfeature. In effect, agreement and movement are triggered by the same feature, hence their co-occurrence. Another existing explanation is that Agree is parameterized to be an upward-probing operation in this language family (Baker 2008). Thus, agreement is only possible after the subject has moved to Spec,TP. The first idea, that movement and agreement are triggered by the same feature, has been called into question in previous literature (see Halpert 2015

for Zulu and Pietraszko 2023b for Ndebele). The Upward Agree approach is largely equivalent to downward Agree followed by movement of the agreement controller. One context in which the predictions of the two analyses diverge is when the goal is *base-generated* in a position c-commanding the ϕ -probe, instead of being moved to that position. Under Upward Agree, such DPs should control agreement all the same because the directionality of Agree is upward. Under downward Agree, such a DP would be inaccessible to the probe because it is not c-commanded by the probe at any point in the derivation.

This prediction can be tested by base-generated topics in Ndebele. In (15-a), the matrix verb is unaccusative and takes a CP complement. Therefore, agreement on matrix T is class 15 (not controlled by any DP). In (15-b), the DP *imota* ‘car’ is in a left-peripheral position in the matrix clause, and it is an antecedent of the object of ‘steal’. The two positions are separated by an island to rule out a movement derivation of the matrix topic. I assume that the object marker in the adjunct CP is agreement with a null argument.

- (15) a. Ku-sobala ukuthi u-danile ngoba ngi-ntshontsh-e imota.
 15s-obvious COMP 2sg-sad because 1sg-steal-PST 9car
 ‘It’s obvious that you are sad because I stole a car.’
- b. Imota_i, ku-sobala ukuthi u-danile ngoba ngi-yi-ntshontsh-ile pro_i.
 9car 15s-obvious COMP 2sg-sad because 1sg-9o-steal-PST 9pro
 ‘As for the car, it’s obvious that you are sad because I stole it.’
- c. *Imota_i, i-sobala ukuthi u-danile ngoba ngi-yi-ntshontsh-ile pro_i.
 9car 9s-obvious COMP 2sg-sad because 1sg-9o-steal-PST 9pro
 ‘As for the car, it’s obvious that you are sad because I stole it.’

As seen from the ungrammaticality of (15-c), a base-generated topic cannot control agreement on T – an incorrect prediction of Upward Agree.⁶

Given this, it is unlikely that upward Agree is the correct explanation of the bi-conditional correlation between agreement and movement. The phasal VoiceP account derives this correlation simply from the standard assumption that ϕ and EPP are both located in T.^{7,8}

⁶ A reviewer suggests that an alternative explanation for (15-c) might appeal to the fact that the topic occupies an A-bar position, while T typically agrees with DPs in A-positions. On this view, the failure of agreement could be characterized as an “improper relation”, by analogy to improper movement. However, characterizing the configuration this way does not in itself constitute an explanation, which would have to come from an analysis of improper relations that could be extended to these data. Most existing accounts of improper relations, however, are narrowly focused on improper *movement* and do not extend to agreement. Two analyses that do so are those of Obata & Epstein 2011 and Safir 2019, but both rely crucially on the involvement of movement. In Obata & Epstein 2011, movement to a lower Spec,TP strips away A-features, rendering them inaccessible to higher A-probes, including ϕ . In Safir’s system, ϕ -features are shielded due to successive cyclic movement through a lower phase edge. Neither of these mechanisms applies in the configuration in (15-c), where no movement occurs. Thus, this configuration does not emerge as an improper relation on either account.

⁷ To be precise, the phasal VoiceP analysis would derive this correlation even if EPP and ϕ were located on different heads, as long as both of those heads were outside of VoiceP.

⁸ It is also worth noting that, under the standard assumption that movement to Spec,TP is triggered by EPP, movement is a downward probing operation. This means that while Upward Agree could in principle apply to ϕ -agreement, in the case of movement the probe (EPP) would c-command its goal. A unified Upward

2.3 Uniform movement and agreement in Aux-V constructions

The phasal VoiceP hypothesis correctly derives the distribution of subject movement and agreement in auxiliary-verb constructions. A well-known property of Bantu languages is the fact that subject agreement morphology can appear on multiple elements in a single clause. In auxiliary-verb constructions such as (16), every verbal element has a subject agreement prefix.

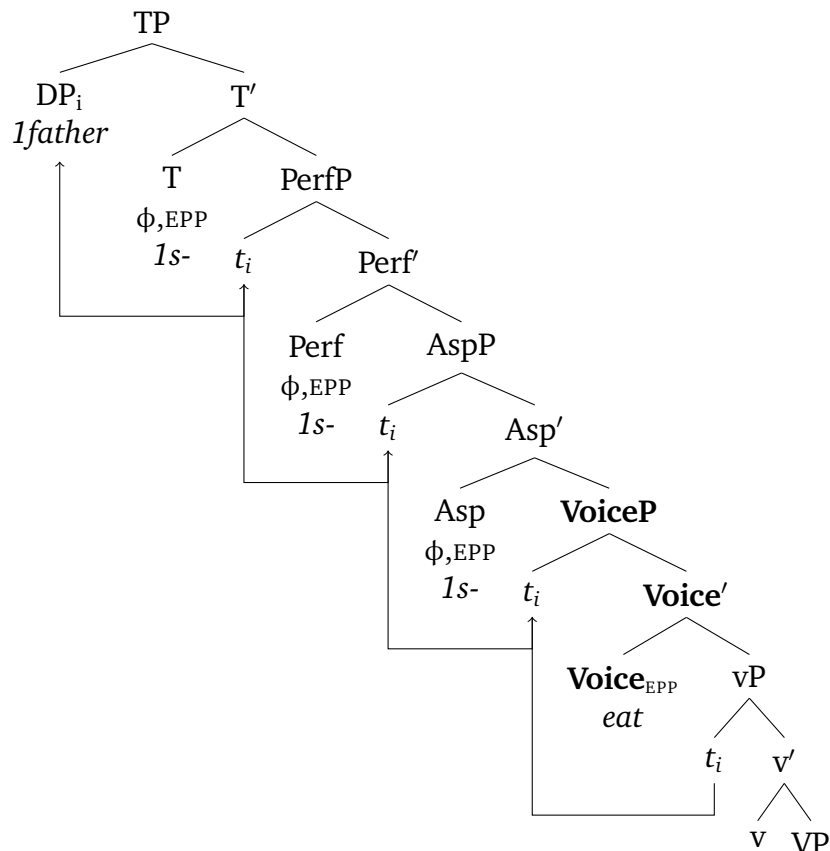
- (16) [TP Ubaba **u**-be [PerfP **e**-se [AspP **e**-si- [VoiceP dla.]]]]
 1father 1s-AUX.PST 1s-AUX.PERF 1s-PROG- eat
 Father had already been eating.

Note that the subject agreement prefix for class 1 has two allomorphs here: /u/ when in T, and /e/ on lower functional heads. This distinction will become relevant in section 3.1.3.

The traditional analysis of Aux-V constructions in Bantu languages is that an agreement probe appears not only on T but also on other functional heads, such as Perf and Asp (i.a. Carstens & Kinyalolo 1989; Carstens 2001; Baker 2008; Pietraszko 2017b). On the assumption that ϕ -probes co-occur with EPP in Bantu languages (Baker 2003, 2008; Collins 2004; Carstens 2005; Pietraszko 2023b), these two features appear on T, Perf, and Asp in the derivation of (16), given in (17). Additionally, if Voice a phase head, it must also have EPP in this derivation to allow the subject to vacate the clause internal phase.

Agree account of ϕ -agreement and movement would therefore additionally require a theory of movement without EPP, where movement is driven by the need to check some feature of the moving DP (see, e.g., Bošković 2007; Bjorkman & Zeijlstra 2019). This is not the standard approach to A-movement in Bantu languages, and one difficulty with adopting it is the lack of a clear candidate for such a feature on the subject. The classic candidate is case, but there is virtually no empirical evidence that case plays a role in movement in most Bantu languages.

(17) The structure of (16)

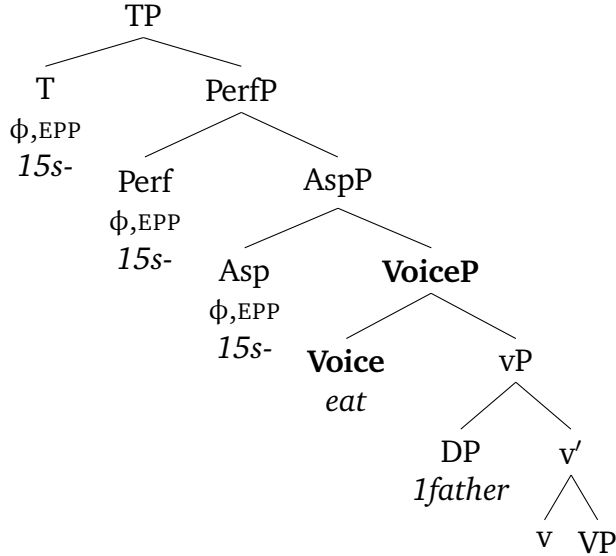


When the subject remains in situ, it follows all verbs and doesn't control agreement on any of them.

- (18) Ku-be ku-se ku-si-dla **ubaba**.
 15s-AUX.PST 15s-AUX.PERF 15s-PROG-eat 1father
 Father had already been eating.

On the phasal VoiceP account, the subject stays in situ whenever Voice lacks EPP. When the subject cannot move to Spec, VoiceP, it is invisible to the EPP features in Asp, Perf and T. It therefore remains inside vP (19). For the same reason, the in-situ subject is invisible to the ϕ -probes in these heads, resulting in class 15 agreement on all of them.

(19) The structure of (18)

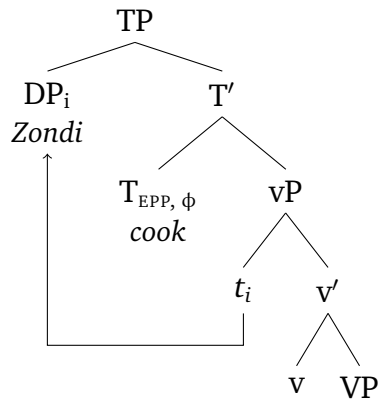


This account makes the following predictions about Aux-V constructions. First, it predicts that all verbal elements must uniformly agree, as in (16), or uniformly fail to agree, as in (18). If the subject moves to Spec,VoiceP, it is visible to all ϕ -probes above Voice, resulting in uniform agreement by all heads. If the subject stays inside the vP, it is visible to none of the ϕ -probes above VoiceP, resulting in uniform default agreement. The examples in (20) further exemplify the all-or-nothing nature of agreement in these constructions. When one head agrees, all others must too. This holds irrespective of whether the subject moves.

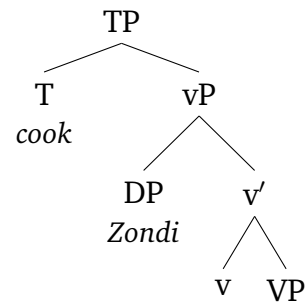
- (20) a. *{Ubaba} **ku-be** **ku-se** **ku-si-dla** {ubaba}.
 1father 1-AUX.PST 15-AUX.PRF 15-PROG-eat 1father
 Father had already been eating.
- b. *{Ubaba} **ku-be** **e-se** **ku-si-dla** {ubaba}.
 1father 15-AUX.PST 1-AUX.PRF 15-PROG-eat 1father
 Father had already been eating.
- c. *{Ubaba} **ku-be** **ku-se** **e-si-dla** {ubaba}.
 1father 15-AUX.PST 15-AUX.PRF 1-PROG-eat 1father
 Father had already been eating.

Relatedly, the second prediction is that the subject can either move all the way to Spec,TP or it does not move at all. The lack of movement results from the lack of EPP in Voice, as already explained for (19). When EPP is present, it attracts the subject to Spec,VoiceP, making it visible to all heads above Voice, including T. Assuming that operations are obligatory when possible, the subject must move to Spec,TP in this derivation. As shown in (21), intermediate landing sites for the subject are indeed impossible, irrespective of the agreement on the functional heads.

- (22) UZondi u-a-phek-a.
1Zondi 1s-PST-cook-FV
'Zondi cooked.'



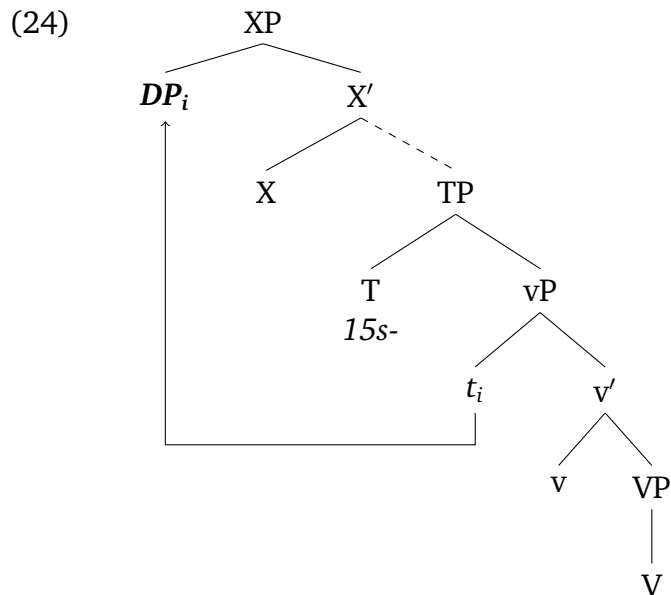
- (23) Ku-a-phek-a uZondi.
15s-PST-cook-FV 1Zondi
'Zondi cooked.'



This analysis faces three problems: i) it overgenerates default agreement, ii) it fails to derive movement and agreement uniformity in Aux-V constructions, and iii) it is directly contradicted by evidence showing that T always has EPP. Subsections 3.1.1-3.1.3 discuss each problem in turn.

3.1.1 Overgeneration of default agreement

The idea that T has an optional [EPP,φ] bundle correctly derives that movement to Spec,TP requires subject agreement, while the lack of movement correlates with the lack of agreement. However, movement to a position higher than Spec,TP should be possible without subject agreement on this account, as this movement has nothing to do with the features of T. [EPP,φ] on T could therefore be missing. This scenario is schematized in (24).



An example of such movement is raising-to-object. In R-to-O constructions, the embed-

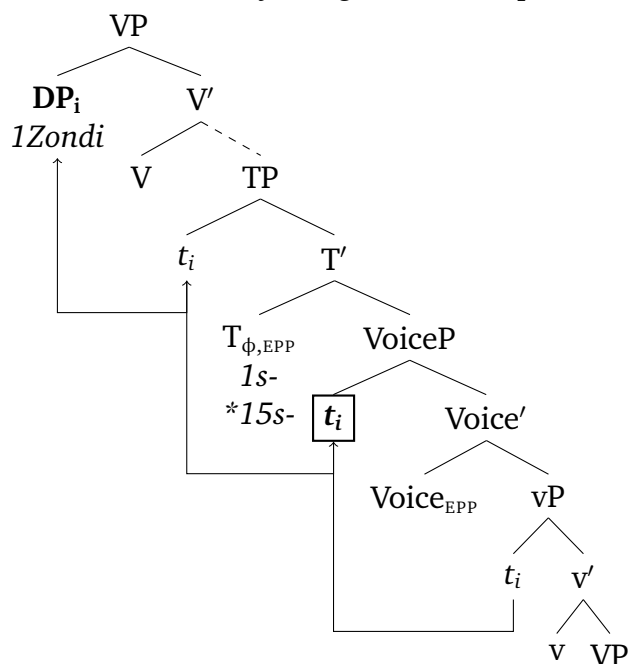
ded subject can surface in three positions: in-situ (25-a), in the embedded Spec,TP (25-b), and in the matrix VP (25-c).

- (25) a. Ngi-funa ukuthi {**ku**/***a**}-pheke **uZondi**.
 1sg-want COMP {15s/*1s}-cook.SBJV 1Zondi
 'I want Zondi to cook.'
- b. Ngi-funa ukuthi **uZondi** {**a**/***ku**}-pheke.
 1sg-want COMP 1Zondi {1s/*15s}-cook.SBJV
 'I want Zondi to cook.'
- c. Ngi-funa **uZondi** ukuthi {**a**/***ku**}-pheke.
 1sg-want 1Zondi COMP {1s/*15s}-cook.SBJV
 'I want Zondi to cook.'

If $[EPP, \phi]$ was optionally absent from T, we would predict the possibility of default agreement (*ku*-) on the embedded T in (25-c), where the embedded subject raises to matrix VP. This would be derived when the matrix verb is merged with a clause whose T lacks $[EPP, \phi]$ (which (25-a) shows to be independently possible). This prediction is incorrect – raising to object requires subject agreement in the embedded clause.

The obligatoriness of embedded subject agreement in (25-c) follows from the present account, in which T always has $[EPP, \phi]$, and VoiceP is a phase. On this account, movement out of VoiceP requires an intermediate step in Spec, VoiceP, a position from which the subject is visible to T. A relation between T and the subject is then inevitable whenever the subject vacates VoiceP. The derivation of (25-c) is given in (26).

- (26) Phasal VoiceP: subject agreement required whenever the subject vacates VoiceP



The phasal VoiceP analysis makes a more general prediction about subject movement and agreement: Movement of the subject to *any* position outside of its local VoiceP should

require subject agreement because a step in Spec,VoiceP is always required for further movement. I am not aware of counterexamples to this prediction.¹¹ For instance, raising to subject behaves the same way as raising to object (27).

- (27) a. Ku-fanele ukuthi {**ku**-/***a**}-pheke uZondi inyama.
 15s-must COMP {15s/*1s}-cook.SBJV 1Zondi 9meat
 ‘Zondi must cook meat.’
- b. Ku-fanele ukuthi uZondi_i {**a**-/***ku**}-pheke *t_i* inyama.
 15s-must COMP 1Zondi {1s/*15s}-cook.SBJV 9meat
 ‘Zondi must cook meat.’
- c. UZondi_i u-fanele ukuthi *t_i* {**a**-/***ku**}-pheke *t_i* inyama.
 1Zondi 1s-must COMP {1s/*15s}-cook.SBJV 9meat
 ‘Zondi must cook meat.’

$\bar{A}$ -movement, such as subject topicalization shown in (28), corroborates the conclusion from A-movement as it too requires subject agreement between T and the moved DP.

- (28) UZondi_i kusasa *t_i* {**u**-/***ku**}-za-pheka inyama.
 1Zondi tomorrow {1/*15}-FUT-cook 9meat
 ‘As for Zondi, tomorrow she will cook meat.’

Obligatory agreement with $\bar{A}$ -extracted arguments is predicted by the phasal VoiceP account, which requires an intermediate step in Spec,VoiceP, a position visible to T.

In sum, the hypothesis that T in Ndebele optionally lacks EPP and ϕ overgenerates default agreement in cases when subject movement terminates in a position other than Spec,TP.

¹¹A reviewer brings up two interesting potential counterexamples. One come from nominal predication, where the subject need not control agreement on the predicate:

- (i) USipho kw-a-ku-ngu-mngane wami
 1Sipho 17S-PST-17S-COP-1friend 1my
 ‘Sipho was my friend.’

Following Buell (2012)’s analysis of the same fact in Zulu, I assume that the subject (USipho) in (i) is base-generated in a topic position. If that’s the case, this DP is not expected to control agreement on T (assuming downward Agree, as I do here).

The second potential counterexample is the fact that class 15 agreement can appear in some cases of coordinated subjects, even if the subject is preverbal. However, I do not take conjunct agreement to be a reliable diagnostic for the workings of Agree, as its surface realization is known to be shaped by numerous non-syntactic factors, including linear proximity, morphological markedness, and processing constraints. Prior work has argued that resolution in such cases may occur post-syntactically (Arregi & Nevins 2012; Bhatt & Walkow 2013), during linearization (Embick & Noyer 2007), or be shaped by processing constraints (Bock & Miller 1991; Wagers et al. 2009). See Lyskawa 2021 for a recent overview and approach.

3.1.2 Non-uniformity of agreement and A-movement in Aux-Vs constructions

The optionally-probing T hypothesis fails to explain the absence of partial movement and agreement in auxiliary-verb constructions, such as (29). As discussed above, a subject in Spec,TP controls agreement on all relevant functional heads (here T and Asp), while an in-situ subject cannot control agreement on either head.

- (29) a. Ubaba u-a-ye e-si-dla.
 1father 1s-PST-AUX 1s-PROG-eat
 ‘Father was eating’
 b. Ku-a-∅ ku-si-dla ubaba.
 15s-PST-AUX 15s-PROG-eat 1father
 ‘Father was eating’

Recall that, on the optional EPP account, (29-a) is derived when both T and Asp have [EPP,φ], while (29-b) is derived when neither T nor Asp have those features.

Given that both T and Asp can optionally lack [EPP,φ], this account predicts that each head can be merged with or without those features independently of the other head. When only Asp is merged with [EPP,φ], we derive (30), where the subject moves only to AspP and controls agreement only on Asp. Conversely, if only T is merged with [EPP,φ], we derive (31), where the subject moves to Spec,TP and controls agreement only on T. The ungrammaticality of both sentences shows that these are incorrect predictions.

- (30) *Ku-a-(ye) ubaba e-si-dla.
 15s-PST-AUX 1father 1s-PROG-eat
 ‘Father was eating’
 (31) *Ubaba u-a-(ye) ku-si-dla.
 1father 1s-PST-AUX 15s-PROG-eat
 ‘Father was eating’

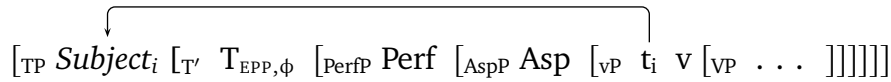
To rule out partial movement and agreement in Aux-V constructions, this account requires a stipulation that all heads in the clausal spine must uniformly be merged with or without [EPP,φ]. In contrast, this uniformity follows as a prediction of the phasal VoiceP account, as explained in section 2.3.

The absence of partial movement in Aux-V constructions has been explained differently by Henderson (2006), who proposes that, in Swahili, intermediate functional heads such as Asp are never merged with EPP. Only T triggers A-movement in this language. For this reason, subject movement can never terminate in any other position than Spec,TP.¹² On the first glance, combining this analysis with the claim that T optionally lacks [EPP,φ] allows us to derive the optionality of subject movement in Ndebele without incorrectly predicting partial movement. When T is merged with [EPP,φ], the subject moves all the

¹²Henderson derives uniform agreement by proposing that lower functional heads copy the φ-features of T. This idea is unlikely to succeed for Ndebele, in which subject agreement is found in reduced clausal complements smaller than TP (see Pietraszko 2017b). Thus, lower functional heads like Asp must be able to agree with the subject DP independently of T.

way to Spec,TP, as shown in (32). When T is merged without those features, the subject stays in situ. These two are the only positions for the subject allowed on this account.

(32) *Only T can bear [EPP, ϕ] $\rightarrow$ either movement to TP or no movement at all*

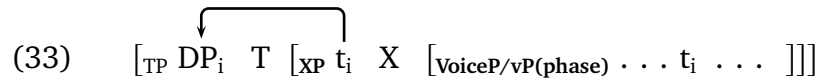


This modification of the optional probing analysis gets rid of only one problem, namely allowing partial movement in Aux-V constructions. It still overgenerates default agreement in raising constructions (section 3.1.1), as it also relies on the optionality of [EPP, ϕ] in T. As we will see in the next section, it is additionally incompatible with the fact that subject movement can terminate in AspP in reduced clauses.

3.1.3 EPP in T is obligatory, not optional

The phasal VoiceP account and the optionally-probing T account differ in the location of the optional EPP feature: On the phasal VoiceP analysis, it's Voice that may lack EPP. On the optionally-probing T account, it is T that lacks EPP in VSO orders. In some general sense, the two analyses are equivalent: The subject stays in situ whenever the relevant EPP is missing.

However, the two analyses make different predictions about movement to TP from a position *outside* of the clause internal phase. Such a configuration is schematized in (33), where the relevant movement step is marked with an arrow.



The subject in Spec,XP has vacated the clause internal phase and thus is visible to T. On the account argued for in the paper, movement from Spec,XP to Spec,TP is obligatory because T always has EPP. In contrast, an analysis that posits optional probing by T predicts that this movement should be optional, as well. In this section, I argue that, once the subject vacates the clause internal phase and lands in an intermediate position, further movement to TP is in fact obligatory. This means that T in this language always has EPP and that the locus of optionality is lower in the structure.

A crucial part of this argument is identifying a projection corresponding to XP in (33). An obvious candidate is AspP, found in Aux-V constructions discussed in the previous section. We have seen that subject movement cannot terminate in Spec,AspP in Ndebele (30). This is consistent with the phasal VoiceP account: the subject must move from AspP to TP because T always has EPP and there is no phasal boundary separating the two positions. However, these facts are also compatible with the modified version of the optionally-probing T account, in which intermediate heads like AspP never have EPP (Henderson 2006). In other words, it is not clear whether the impossibility of partial movement is due to an obligatory EPP in T or due to systematic absence of EPP on intermediate heads.

This dilemma is resolved by evidence showing that Asp does in fact have EPP. This can be seen in reduced clausal adjuncts, which lack the TP layer. An example is given in (34), whose literal translation better reflects the reduced status of the adjunct clause.

- (34) Ngi-khulume, [(**ukuthi*) ubaba e-si-dla].
 1sg-speak.PST (**COMP*) 1father 1PTCP-PROG-eat.
 ‘I spoke while father was eating.’ (Lit. I spoke, father eating.)

The first indication that the adjunct clause in (34) is reduced is the impossibility of an overt complementizer. Further evidence reveals that it is also smaller than a TP. First, the verb in the reduced clause has the same as the form of the progressive participle in a progressive compound tense (35).

- (35) Ubaba u-Ø-be e-si-dla.
 1father 1-PST-AUX 1PTCP-PROG-eat
 ‘Father was eating.’

Recall that class 1 agreement exhibits allomorphy depending on the head it’s located on: the allomorph *u-* appears on T, while the allomorph *e-* appears on lower heads, such as Asp or Perf (16)-(17). Importantly, the agreement prefix in reduced adjunct clauses must take the participial allomorph found on the lower functional heads (36).

- (36) Ngi-khulume, ubaba e/**u*-si-dla.
 1sg-speak.PST 1father 1PTCP/**1*-PROG-eat.
 ‘I spoke while father was eating.’ (Lit. I spoke, father eating.)

Second, no tense marking is possible in the reduced clausal adjunct:

- (37) *Ngi-khulume, ubaba e-{*a/za/ya*}-si-dla.
 1sg-speak.PST 1father 1-{PST/FUT/PRS-PROG}-eat.
 ‘I spoke while father was eating.’

Finally, a TP can be diagnosed by the possibility of containing auxiliary verbs. In progressive tenses, such as (35) above, the lexical verb is inflected for aspect, while T attaches to the default auxiliary *be*. (38) shows that auxiliaries can occur in non-finite TPs as well.

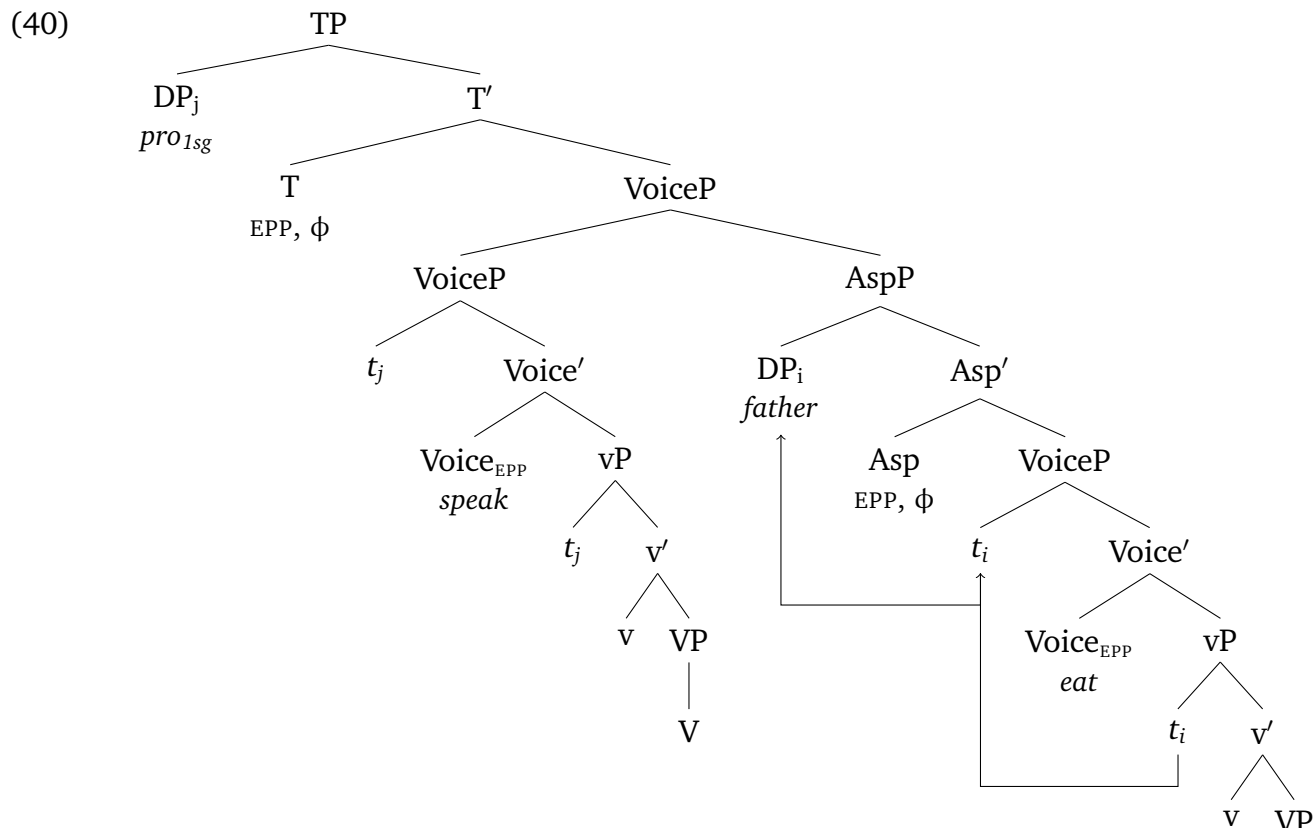
- (38) Ubaba u-funa uku-**be** e-si-dla.
 1father 1-want INF-AUX 1PTCP-PROG-eat
 ‘Father wants to be eating.’

There are various approaches to the syntax of compound tenses, but they all seem to converge on the general idea that the presence of the auxiliary is triggered by the need to realize tense inflection – something the progressive participle cannot do, for reasons that vary across analyses. This dependency on T has been formalized either as T selecting an auxiliary (i.a. Ross 1969; Déchaine 1995; Tenny 1987; Adger 2003; Pietraszko 2017b, 2023a) or as the auxiliary being inserted in the head position realized by the tense affix – typically T in most accounts (i.a. Arregi 2000; Embick 2000; Cowper 2010; Bjorkman 2011; Calabrese 2019; Fenger 2020). On either view, the presence of a TP layer implies the potential for an auxiliary.

If adjunct clauses are as large as TP, we would expect them to permit compound tenses. This is, however, not the case:

- (39) *Ngi-khulume, ubaba e/u-be e-si-dla.
 1sg-speak.PST 1father 1PTCP/1-AUX 1PTCP-PROG-eat.

All the evidence above points to the same conclusion, namely that the adjunct clause in (34) is only as large as AspP. The derivation of (34) is given in (40).



Despite being smaller than a TP, the reduced adjunct clause can have a preverbal agreeing subject. Since its highest projection is AspP, subject movement and agreement must be triggered by Asp, rather than T. The participial form of the subject agreement prefix is an especially strong indicator that the agreeing head is Asp, not T. This means that Asp in Ndebele is a head that has (or at least can have) both EPP and ϕ .

By ruling out the hypothesis that Asp never has EPP, we have identified AspP as a projection that corresponds to XP in (33) – a projection above the clause internal phase but below T, to which the subject can move. We can therefore use Aux-V constructions to choose between the VoiceP analysis and the optionally-probing T analysis based on the different predictions they make about the movement indicated in (33).

If subject movement can terminate in AspP in reduced clauses (41-a), we must ask why it cannot terminate in the same AspP in Aux-V constructions (41-b).

- (41) a. Ngi-khulume, [_{AspP} ubaba e-si-dla].
 1sg-speak.PST 1father 1-PROG-eat.
 ‘I spoke while father was eating.’

- b. *Ku-Ø-be [AspP ubaba e-si-dla.]
 15-PST-AUX 1father 1-PROG-eat
 ‘Father was eating’

The obvious answer is: Subject movement can terminate in AspP only when AspP is the highest projection in the clausal spine. This is the case in the reduced clause (41-a), but not in the Aux-V construction (41-b). The latter is an instantiation of (33), where AspP is the intermediate XP. What we learn, then, is that when TP is present, movement to Spec,TP is obligatory. This means that T always has EPP in Ndebele, directly contradicting the claim that EPP in T is optional. The phasal VoiceP account makes the correct prediction: Since T always has EPP, a subject that has vacated the clause internal phase must move all the way to Spec,TP. Movement terminating in AspP is only possible when T is not merged at all, as in (41-a).

For completeness, note that subjects can stay in-situ in reduced clauses, as well. As expected, agreement on Asp in such cases takes the default form *ku*:

- (42) Ngi-khulume, [AspP ku-si-dla ubaba].
 1sg-speak.PST 15-PROG-eat 1father.
 ‘I spoke while father was eating.’

Identifying AspP as a possible landing site for subject movement is crucial in revealing the locus of optionality in subject movement: It is not the movement *to* TP that’s optional, but movement *from* the base position. Distinguishing these two is impossible if movement from the base position to TP occurs in one step. An intermediate position, such as Asp, reveals that the locus of optionality is below Asp:

- (43)
$$\begin{array}{c} \text{obligatory} \quad \text{optional} \\ \downarrow \quad \downarrow \quad \downarrow \\ [_{TP} DP_i \ T \ [_{AspP} DP_i \ Asp \ \dots \ [_{VP} DP_i \ v \ \dots \]]] \end{array}$$

This result is in line with the phasal VoiceP account, which attributes the optionality of subject movement to an optional EPP feature on phasal Voice.

This completes our discussion of the first probe-based account, which attempts to derive the SV-VS alternation from an optional EPP in T. The second probe-based account, discussed in the next section, proposes that probing by T is blocked by an expletive subject in VS orders.

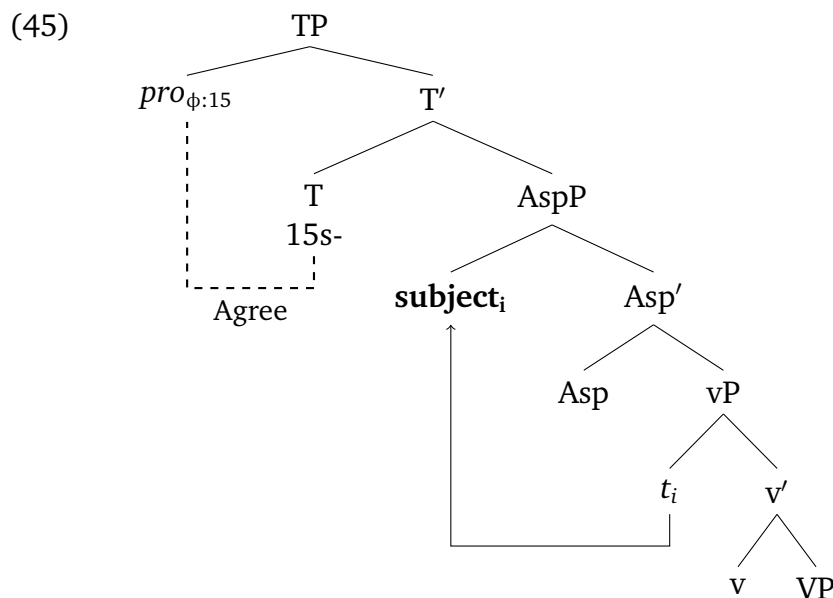
3.2 Probing blocked by expletive *pro*

Buell 2005, 2007; Halpert 2015 advocate an analysis of the SV-VS alternation in Zulu according to which an expletive *pro* of class 17¹³ can be inserted in Spec,TP, bleeding movement of the logical subject to this position. Insertion of the expletive is freely available, as stated in (44), from Halpert 2015.

¹³Class 17 and class 15 are indistinguishable in both Ndebele and Zulu. The choice of a specific gloss tends to depend on traditional conventions, such as using class 17 for locative constructions and class 15 for infinitives. Accordingly, Buell’s and Halpert’s class 17 agreement corresponds to my class 15 agreement in VS clauses.

- (44) *Two ways to satisfy the EPP in Zulu* (Halpert 2015:257)
- Insert *pro_{expl}* directly in Spec,TP
 - Search for an argument of the verb and move it to Spec,TP

Adopting this proposal for Ndebele makes an incorrect prediction for Aux-V constructions. If the expletive *pro* is inserted directly in Spec,TP, we predict that subject movement could terminate in Spec,AspP, incorrectly predicting partial movement (45).



We also predict non-uniform subject agreement: T should agree with the expletive *pro* (class 15) and Asp should be able to agree with the logical subject. This outcome is ungrammatical, as we saw in the previous section (41-b). A way to amend this analysis is to posit that the expletive is inserted low (as has been argued for the English expletive *there*; see i.a. Bowers 2002; Richards & Biberauer 2008; Deal 2009; Harwood 2015b). Specifically, the expletive would have to be inserted in the lowest functional projection whose head has a subject agreement probe. Even with this modification, this account would require a further stipulation: only the lowest head with EPP can have the option of merge an expletive instead of moving the subject. Higher heads cannot have this option – otherwise, we would predict partial movement and agreement.

If we adopt this modification, however, the expletive *pro* analysis successfully derives the uniform pattern of movement and agreement observed in auxiliary-verb constructions. On this view, the expletive *pro* is merged low in the structure, but still higher than the logical subject. As a result, it would serve as the closest goal for all inflectional heads bearing ϕ and EPP.

Another correct prediction of the modified expletive *pro* account is that the logical subject could not undergo raising to object in the presence of an expletive *pro*, again due to DP minimality. As shown in (25-c), repeated below in (46), this is manifested by the impossibility of default agreement when the logical subject raises to matrix object:

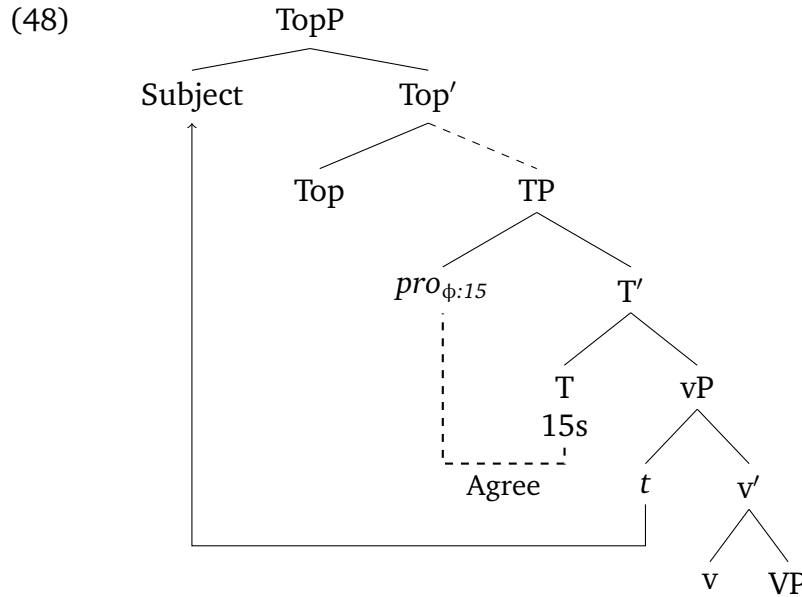
- (46) Ngi-funa **uZondi** ukuthi {**a/*ku**}-pheke.
 1sg-want 1Zondi COMP {1/*15}-cook.SBJV
 'I want Zondi to cook.'

On the expletive *pro* account, class 15 agreement indicates that embedded Spec,TP is occupied by expletive *pro*, which blocks raising of the logical subject. Instead, this account predicts that the expletive itself may undergo raising to object, as expletives are able to do crosslinguistically. Since the expletive here is null, this movement would be string vacuous. One potential diagnostic is matrix object agreement: when a *pro* raises to object, it triggers object agreement in the matrix clause, as seen in (47-a). However, the hypothetical expletive *pro* fails to do so (47-b).

- (47) a. Ngi-**ba**-funa ***pro*_i** [_{CP} ukuthi ***t*_i** ba-pheke.]
 1sg-**2o**-want 2pro COMP 2-cook.SBJV
 'I want them(cl.2) to cook.'
- b. *Ngi-**ku**-funa ***pro*_i** [_{CP} ukuthi ***t*_i** ku-pheke abafana.]
 1sg-**15o**-want 15pro COMP 15-cook.SBJV 2boys
 'I want the boys to cook.'

While this initially looks like an argument against the expletive *pro*, it is worth noting that object agreement in Ndebele is restricted to topical/discourse-given DPs, such as null pronouns. Given this, the contrast in (47) may be attributed to the lack of semantic content of the expletive *pro*, which presumably makes it incompatible with topicality.

In this respect, the expletive *pro* account remains compatible with the raising-to-object facts. Where it fails, however, is in its predictions for A-bar movement: while expletive *pro* is expected to block A-movement, it is not expected to block A-bar movement. Specifically, this account incorrectly predicts that the logical subject could be topicalized across the expletive *pro*. The concrete derivation allowed by this account is shown in (48), where T agrees with the expletive *pro* (the closest DP). This leads to class 15 agreement on T and movement of the *pro* to Spec,TP. Subsequently, the logical subject undergoes topicalization across the expletive subject. As we saw in (28), repeated below in (49), class 15 default agreement is impossible when the subject is topicalized.



- (49) UZondi_i kusasa {u/*ku}-za-pheka inyama.
 1Zondi tomorrow {1s/*15s}-FUT-cook 9meat
 ‘As for Zondi, tomorrow she will cook meat.’

A reviewer suggests the possibility that topicalization in Ndebele is subject to DP minimality, as has been observed in some other languages – most notably in the Austronesian family (see, e.g., Aldridge 2004, 2008a; Rackowski & Richards 2005; Erlewine & Lim 2023; Branan & Erlewine 2024). If this were the case, A-bar extraction would be restricted to the highest DP, deriving the impossibility of subject topicalization across an expletive *pro*. This analysis cannot be maintained, however, simply because A-bar movement, including topicalization, is not restricted to the highest DP in Ndebele. For example, the direct object can be topicalized across both a subject and an indirect object, as shown in (50).

- (50) **Imali_i** *pro* ngi-(yi)-nik-e [_{VP} abafana **t_i**].
 9money *pro*.1SG 1SG-(9o)-gave-PST 2boys
 ‘The money, I gave (it) to the boys’

Topicalization of the theme across two higher arguments strongly indicates that this movement type is not subject to DP minimality in Ndebele. The only way of maintaining this hypothesis would be to propose that the movement in (50) involves two steps of leapfrogging: one around the indirect object and one around the subject. This is schematized in (51).

- (51) A leapfrogging derivation of (50)
- $$[\text{TopP } \text{DP}_{DO} \text{ Top } [\text{TP } t [\text{T}' \text{ DP}_{Subj} \text{ T} \dots [\text{ApplP } t [\text{Appl}' \text{ DP}_{IO} [\text{VP } \text{V } t]]]]]]]$$

The leapfrogging proposal is untenable for three reasons. First, the indirect object always counts as the higher goal for the object agreement probe (which is located in Voice,

see Pietraszko 2023b). This can only be seen when both objects are discourse given, as only discourse given DPs are eligible for object agreement in this language. In (52), both object are a *pro*, and thus are both eligible. Allowing the theme to leapfrog around the indirect object, as in (51), incorrectly predicts that the theme could control object agreement in (52).

- (52) *pro* *ngi*-{**ba**/***yi**}-*nik-ile* *pro*_{IO} *pro*_{DO}.
 PRON.1SG 1SG-{**2o**/***9o**}-gave-PST PRON.2 PRON.9
 ‘I gave it(cl.9) to them(cl.2).’

The same is observed in cases of direct object topicalization: When the indirect object is a *pro*, and thus an eligible goal, agreement with the topicalized direct object is impossible:

- (53) **Imali**_i *pro* *ngi*-{**ba**/***yi**}-*nik-ile* *pro* **t_i**. C.f. (50)
 9money PRON.1SG 1SG-{**2o**/***9o**}-gave-PST PRON.2
 ‘The money, I gave (it) to them.’

Thus, the purported leapfrogging through ApplP is likewise unavailable as an intermediate step of topicalization.

Second, if leapfrogging around the subject in Spec,TP is allowed, as in (51), we predict that the logical subject in (48) should be allowed to leapfrog around the expletive *pro*. This would produce the ungrammatical version of (49) with class 15 subject agreement.

And finally, this step of leapfrogging would allow the logical subject to raise to matrix object across an expletive *pro*. This would lead to the ungrammatical version of (46) with class 15 agreement on embedded T.

In sum, the idea that topicalization in Ndebele has the locality of A-movement makes incorrect predictions, regardless of whether leapfrogging is assumed to be possible. Instead, topicalization in this language behaves like traditional A-bar movement, i.e. movement that can cross other DPs. Given this conclusion, the impossibility of subject topicalization across an expletive in *pro* (49) remains an incorrect prediction of the expletive *pro* account – A-bar movement should not be blocked by a DP.

This prediction extends to object topicalization: Just like subjects, objects should be free to undergo topicalization across an expletive *pro*. For this reason, the expletive *pro* account predicts (54) to be acceptable, contrary to fact.

- (54) ***Inyama**_i *ku*-*phek-e* [_{VP} *umama* **t_i**].
 9meat 15-cook-PST 1mother
 ‘The meat, mother cooked (it).’

If Spec,TP in (54) is occupied by an expletive *pro*, the direct object is attempting to cross two DPs here: an overt DP inside of vP and a *pro* in Spec,TP. The same set of interveners is successfully crossed in (50), which means that the ungrammaticality of (54) is not a case of intervention.

Let us consider how the phasal VoiceP analysis might account for (54). First, note that (50) and (54) together reveal the following generalization:

- (55) If the subject cannot move out of VoiceP, neither can the object.

Recall that when the subject remains in situ, this indicates that Voice was merged without an EPP feature. The absence of EPP, then, appears to preclude object movement as well.¹⁴

One way to implement this connection is to propose that the same feature allows both subject and object movement out of VoiceP. This feature would be capable of probing multiple times, similar to the Edge Feature (Chomsky 2000) or an “insatiable” probe (Deal 2015). On this hypothesis, the subject and object would both be eligible goals for EPP in Voice. This means that the subject, being structurally higher, will always be attracted to Spec,VoiceP before the object, deriving the entailment in (55).¹⁵

An alternative approach that also links object movement to subject movement is offered by Pietraszko (2023b), who observes that object right-dislocation in Ndebele is only possible if the subject has moved out of VoiceP. Building on the idea that VoiceP is a phase, Pietraszko proposes that this dependency follows from a well-attested generalization in this language family: ϕ -probes occur only on heads that also bear an EPP feature (Baker

¹⁴This generalization predicts that long-distance topicalization should be blocked if the matrix subject remains in situ. This is a correct prediction: While long-distance topicalization is allowed in Ndebele (i), it requires the matrix subject to move out of VoiceP (ii).

- (i) USipho_i *pro* ba-cabanga [_{CP} ukuthi nginike *t_i* imali.]
 1Sipho 2.pro 2-think COMP 1sg-gave money
 ‘Sipho, they think that I gave him money.’
- (ii) *USipho_i ku-cabanga [_{VP} abafana [_{CP} ukuthi nginike *t_i* imali.]]
 1Sipho 15-think 2boys COMP 1sg-gave money
 ‘Sipho, the boys think that I gave him money.’

Another possible prediction of this generalization is that XPs originating outside of VoiceP should be extractable in VS sentences, as their extraction does not require EPP in Voice. Buell (2011) argues that in Zulu, the *wh*-phrase *ngani* ‘why’ originates outside vP, as evidenced by the fact that it follows the disjoint form of the verb rather than the conjoint. Testing whether a similar pattern holds in Ndebele must be left to future research, with several issues needing to be addressed in the process. First, it must be established whether the conjoint domain in Ndebele is in fact contained within VoiceP; only then can the conjoint-disjoint alternation be used to diagnose XPs generated outside of VoiceP. Second, recall that the subject in VS sentences is in focus, which may interact with *wh*-movement – another type of focus construction. This complication does not arise for topicalization, which can (and indeed must) co-occur with a focused XP in the main clause. Finally, there is the possibility that A-bar movement of DPs differs from that of other categories. The *wh*-word *ngani* is likely a non-DP, given its morphological shape, so it will be important to first investigate the fronting properties of non-DPs and how they interact with A-movement of the subject. All these interesting questions must remain for future work.

¹⁵On this type of analysis, EPP in Voice would be necessary for object topicalization, but crucially, it couldn’t be sufficient due to the fact that the object need not move out of VoiceP whenever the subject does. One way to reconcile this with insatiable EPP would be to say that intermediate movement though VoiceP is in some sense dependent on/licensed by the presence of a higher probe licensing a final landing site (e.g. T for subjects, a topicalization probe in the left periphery for objects). Cases where only the subject moves would then be those in which no topicalization probe is merged. Another possibility is to abandon the idea of an insatiable EPP and instead allow Voice to host multiple EPP features, each of which can attract only one goal. On this view, subject movement would always be triggered by the first EPP, while object movement would require an additional EPP (which would have to be optional) deriving the generalization in (55). (This is essentially Pietraszko’s 2023 analysis of symmetrical passives of ditransitives in Ndebele.) I leave this question open for future work.

2003, 2008; Collins 2004; Carstens 2005). Assuming that object agreement is a property of Voice (see Pietraszko for evidence), and ϕ in Voice entails EPP, object dislocation indirectly requires EPP in Voice. In other words, i) object dislocation requires ϕ in Voice, ii) ϕ in Voice entails EPP in Voice, iii) and EPP in Voice moves the subject out of VoiceP.

Both analyses are compatible with the current claim that VoiceP is a phase, as they both depend on this assumption. However, Pietraszko's account predicts that object agreement should always accompany movement of the object out of VoiceP, yet this is not borne out for object topicalization, which can occur without object agreement (50). To extend Pietraszko's proposal to topicalization would require positing null object agreement in such cases. In contrast, the single-feature hypothesis sketched above offers a simpler account, as it can uniformly explain both topicalization and right-dislocation. I tentatively suggest that this hypothesis is on the right track, though its full implementation and empirical evaluation lie beyond the scope of this paper. To comment briefly on the nature of this feature, it would have to be a feature that can feed both A and A-bar movement, as it feeds both subject movement to Spec,TP and topicalization. Following the generalization that A-bar movement cannot feed A-movement (Chomsky 1973, 1977, 1981; May 1979), the movement feature in Voice could not be an A-bar feature. Instead, it's possible that Voice in Ndebele (though not necessarily in other languages) has only an A-movement feature. Alternatively, it would be worth exploring the idea that the movement feature in Voice is underspecified for this distinction and so it does not interact with the improper movement generalization. Both implementations have interesting theoretical implications and empirical predictions, which I must leave for future work.^{16,17}

This section examined two analyses of VSO orders in Ndebele that both rely on the lack

¹⁶A question that arises for both analyses (all-A movement or underspecified movement) is how to handle cases of A-bar opacity, where both the subject and an object move through Spec,VoiceP (as would be the case in object topicalization). The question is what ensures that T targets the subject rather than the object in such configurations. Since the object would not be in an A-bar position on either account, existing analyses of improper movement would not be applicable here. One possible solution would be that the object tucks in under the subject in VoiceP, though other options may be available depending on how the analysis is ultimately developed. I flag this as an open issue that a fuller implementation of the ideas sketched here.

¹⁷In this paper, I have used topicalization to illustrate A-bar movement. The choice is motivated by the fact that other potential candidates are not straightforwardly movement constructions. Focus clefts (i) and wh-questions (ii) are island-insensitive, even with respect to robust crosslinguistic islands such as reason-CPs. In fact, even topicalization is island insensitive if the gap is marked with an object marker (iii), which likely agrees with a null argument in the base position. In contrast, topicalization without an object marker is sensitive to islands (iv). These facts indicate that the constructions in (i)–(iii) either do not involve movement or allow a non-movement derivation. Either way, they do not provide a clear basis for investigating the behavior of A-bar movement. For this reason, the discussion here is restricted to topicalization without an object marker, where the confound does not arise.

- (i) Leli yilo **ikhhekhe** e-ngi-buy-ile-yo ngoba uZondi u-be-e-**li**-pheka.
 DEM.5 COP.5 5cake REL-1SG.S-come-PST-REL because 1Zondi 1s-AUX-1s-5o-cook
 'This is the cake that I came because Zondi was making it.'
- (ii) Yi-**ni** e-li-suk-ile-yo ngoba uZondi ka-**yi**-phek-anga?
 COP-9what REL-2PL.S-leave-PST-REL because 1Zondi 1s.NEG-9o-cook-NEG
 'What did you leave because Zondi didn't cook it?'

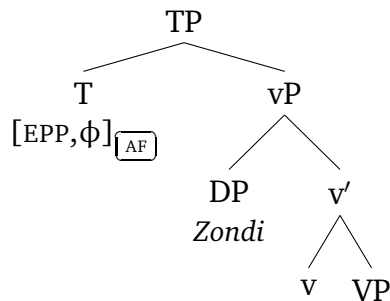
of probing for the logical subject by T. On the analysis discussed in section 3.1, T fails to probe the subject due to not having the relevant probes ([EPP, ϕ]). According to the other analysis, probing is blocked by an expletive *pro* (§3.2). Both accounts were shown to overgenerate class 15 agreement: the expletive *pro* incorrectly predicts class 15 subject agreement in topicalization, while the first account overgenerates class 15 agreement in both topicalization and raising. Additionally, the claim that T optionally lacks EPP was shown to be incorrect by the fact that movement from AspP to TP is obligatory. These results lead to the conclusion that VSO clauses in Ndebele are not a problem of probing. The probe features are always present in T and the search is not blocked by another DP.

4 Lack of a suitable goal

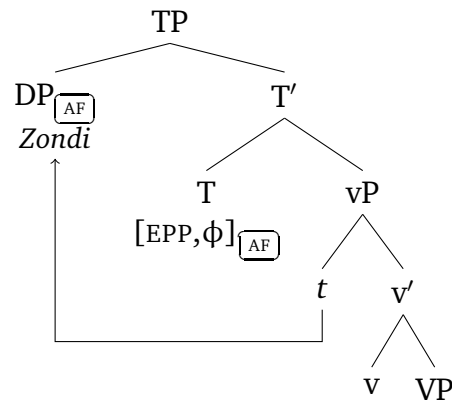
The previous section argued that the absence of subject movement to Spec,TP in Ndebele is not a failure of probing. The other possibility is that the problem lies with the goal. An analysis in this vein was proposed by Zeller (2008; 2015) to account for VSO constructions in Zulu. In this section, I apply Zeller’s analysis to Ndebele and argue that it cannot successfully derive the SV-VS alternation in this language. We will see that the subject can stay in vP despite being independently shown to be a matching goal for T.

Zeller (2008, 2015) proposes that T in Zulu always has EPP and ϕ . However, T is relativized to attract only a subset of DPs, namely ones with the Antifocus (AF) feature. When the subject lacks AF, it is not a matching goal for T. Therefore it neither moves to Spec,TP nor controls agreement on T (56). Subject movement and agreement occur always and only when the subject bears AF (57).

- (56) Ku-a-phek-a uZondi.
 15s-PST-cook-FV 1Zondi
 ‘Zondi cooked.’



- (57) UZondi u-a-phek-a.
 1Zondi 1s-PST-cook-FV
 ‘Zondi cooked.’



- (iii) **Imota**, ngi-biz-e amapolisa ngoba ba-yi-ntshontsh-ile.
 9car 2SG.S-call-PST 6police because 2s-9o-steal-PST
 The car, I called the police because they stole it.

- (iv) ***Imota**, ngi-biz-e amapolisa ngoba ba-ntshontsh-ile.
 9car 2SG.S-call-PST 6police because 2s-steal-PST
 The car, I called the police because they stole it.

The appeal of this account is that it additionally attempts to explain the information structural properties of SV and VS orders: Subjects in Spec,TP cannot be narrowly focused, as shown by the Ndebele sentence in (58-a). This restriction does not hold for in-situ subjects (58-b).

- (58) a. *[UZondi kuphela]_{Foc} u-a-phek-a.
 1Zondi only 1s-PST-cook-FV
 ‘Only Zondi cooked.’
 b. Ku-a-phek-a [uZondi kuphela]_{Foc}.
 15s-PST-cook-FV 1Zondi only
 ‘Only Zondi cooked.’

The impossibility of focus on the DP in Spec,TP is captured by the proposal that T probes only for DPs with the Antifocus feature, which is incompatible with a focus interpretation.

However, as has been noted in Pietraszko (2021), the restriction we observe in (58-a) is not due to movement to Spec,TP per se. Rather, it is a property of preverbal subjects in root and root-like clauses. Movement to Spec,TP in subjunctive and relative clauses is compatible with focus (59).

- (59) a. Ngi-funa [ukuthi **u-Sipho kuphela** a-buye].
 1SG-want.CNJ COMP A-1Sipho only 1S-come.SBJV
 ‘I want only Sipho to come.’
 b. Ngi-zwe ingoma [**u-Sipho kuphela** a-yi-hlabelileyo].
 1SG-heard.CNJ 9song A-1Sipho only 1S-9O-sing.PST.REL
 ‘I heard the song that only Sipho sang.’ (Pietraszko 2021:591)

The facts in (59) show that subject movement and agreement are not directly correlated with focus and thus they are likely not driven by a focus-related feature, such as AF.

One might ask whether there could be a different feature that occurs optionally on DPs and that regulates movement and agreement through relativized probing. In the absence of systematic semantic differences between Spec,TP and the base position (e.g. both are compatible with focused DPs, definites, indefinites, quantifiers, and wh-phrases), such a feature is unlikely to have an interpretation. Nonetheless, there remains a possibility of a purely syntactic feature that serves the role of AF is Zeller’s analysis of subject movement. In the remainder of this section, I argue that there is no additional feature of DPs that drives movement and agreement in Ndebele.

Let us start from the observation that a relativized probing analysis can derive the impossibility of partial movement in Aux-V constructions (60). This can be accomplished if we posit that all functional heads in the clausal spine, including Perf and Asp, are relativized to the same optional feature of DPs as T. Let us refer to this feature as MF for “movement feature”. (Agreement in parasitic on movement in Zeller’s analysis.) Movement to Spec,AspP, for instance, would entail that the subject has MF. This, in turn, entails that the subject is a matching goal for T, as well. Movement all the way to Spec,TP must therefore take place, ruling out partial movement to Spec,AspP.

- (60) *_{[TP} Ku-be _{[AspP} **ubaba**_{MF} e-si- _{[vP} **t dla**]]]
 15-AUX.PST 1father 1-PROG eat
 ‘Father was eating’.

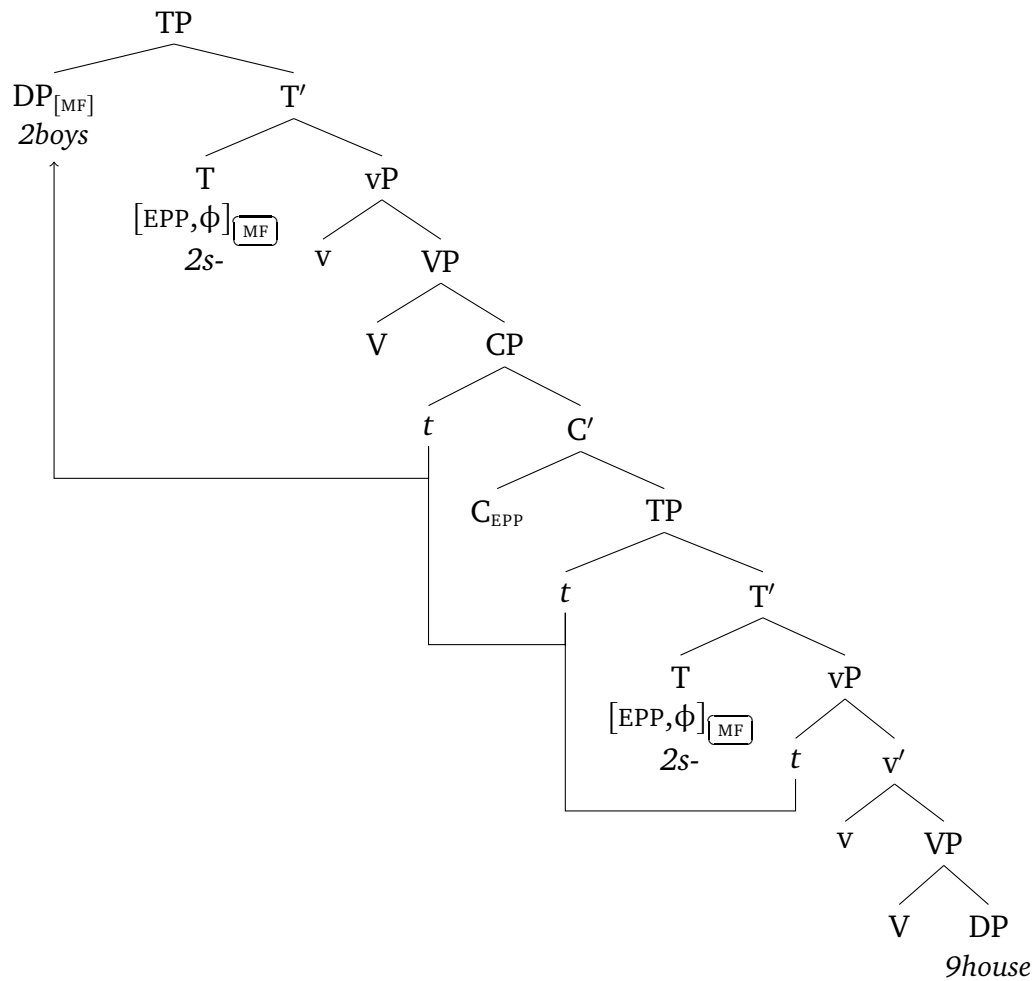
The same relativized probing analysis can be applied to cross-clausal A-movement, such as hyperraising, illustrated below. The matrix verb *jaye*, which roughly translates as ‘usually’, is an unaccusative verb selecting a subjunctive CP. The derived nature of the matrix subject in (61-a) is evidenced by i) the semantic equivalence of the raised and unraised counterparts in (61), ii) the active-passive synonymy in (62), and iii) the preservation of idiomatic meaning (63).

- (61) a. **Abafana**_i ba-jaye [CP ukuthi t_i ba-pende t_i indlu.]
 2boys 2-usually COMP 2-paint.SBJV 9house
 ‘The boys usually paint the house.’
 ≈
 b. Ku-jaye [CP ukuthi **abafana**_i ba-pende t_i indlu.]
 15-usually COMP 2boys 2-paint.SBJV 9house
 ‘The boys usually paint the house.’
- (62) a. **Abafana**_i ba-jaye [CP ukuthi t_i ba-pende t_i indlu.]
 2boys 2-usually COMP 2-paint.SBJV 9house
 ‘The boys usually paint the house.’
 ≈
 b. **Indlu**_i i-jaye [CP ukuthi t_i i-pendwe t_i ng-abafana.]
 9house 9-usually COMP 9-paint.PSV.SBJV by-2boys
 ‘The house is usually painted by the boys.’
- (63) a. **Isigogo** si-a-goqwa sisemanzi.
 7leather 7-PST-fold.PSV wet.PTCP
 Lit: ‘Leather was folded while still wet’
Idiom: ‘It was done at the right time.’ ✓
- b. **Isigogo**_i si-jaye [CP ukuthi t_i si-goqwe t_i sisemanzi.] *Raising*
 7leather 7-usually COMP 7-fold.PSV.SBJV wet.PTCP
 Lit: ‘Leather is usually folded while still wet’
Idiom: ‘Something/it is usually done at the right time.’ ✓
- c. **Isigogo** si-zama [CP ukuthi si-goqwe sisemanzi.] *Control*
 7leather 7-try COMP 7-fold.PSV.SBJV wet.PTCP
 Lit: ‘#Leather tried to be folded while still wet’
Idiom ✗

The embedded subject in these hyperraising sentences controls agreement on the embedded T, which indicates that it has the movement feature that allows it to interact with T. The same feature makes the embedded subject a matching goal for matrix T, resulting in raising to matrix Spec,TP. The derivation for (61-a) is given in (64). (I assume that A-movement crosses the CP phase successive cyclically (see e.g. Fong 2019), but other

analyses are also compatible with the present discussion.)

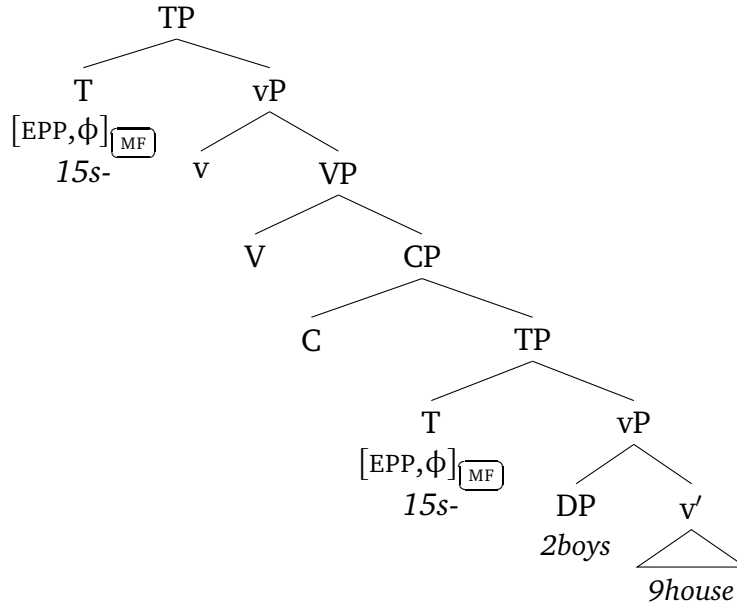
- (64) Movement to matrix TP → the subject has [MF]



In contrast, when the embedded subject lacks [MF], it cannot move at all and thus remains in its base position in the embedded clause. Default agreement appears on both the matrix and the embedded T (65)-(66).

- (65) **Ku**-yajele [_{CP} ukuthi **ku**-pende **abafana** indlu.]
 15s-usually COMP 15s-paint.SBJV 2boys 9house
 'Usually, the boys paint the house.'

- (66) No movement of the subject → the subject does not have [MF]



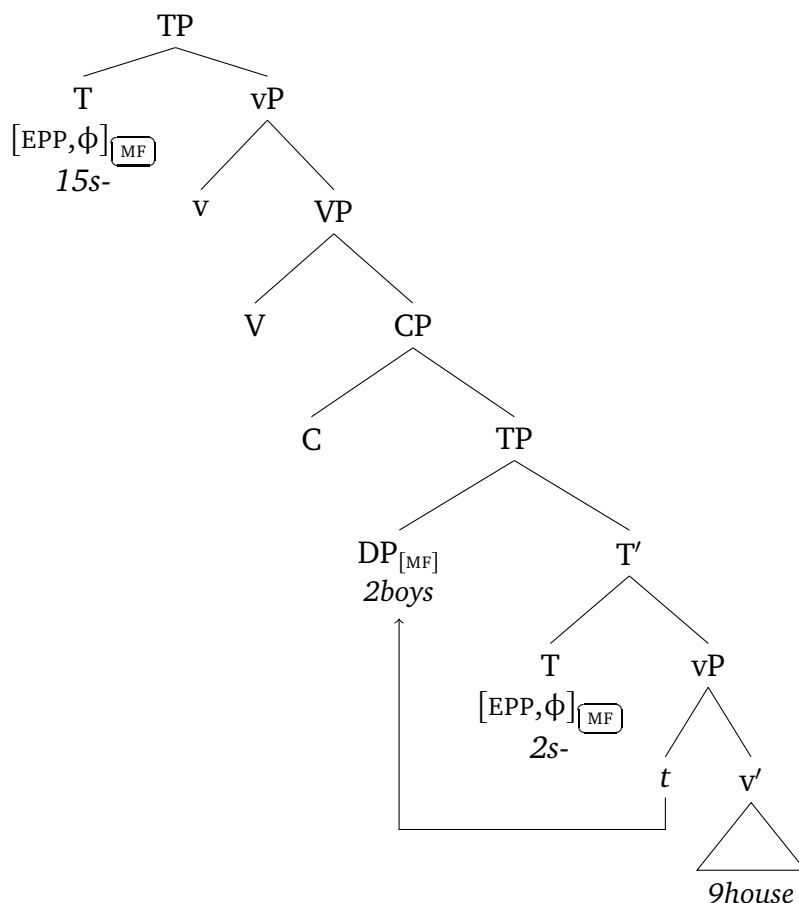
Matrix Spec,TP and the base position are the only two subject positions predicted by the relativized probing account. Recall from the discussion above that partial movement is generally disallowed on this analysis – a DP either does or does not have the movement feature. Contrary to this prediction, there are two other positions in which the subject can surface in raising constructions.

First, subject movement can terminate in the embedded Spec,TP (67).

- (67) Ku-jaye le [CP ukuthi **abafana**_i ba-pende *t_i* indlu.]
 15s-usually COMP 2boys 2s-paint.SBJV 9house
 ‘The boys usually paint the house.’

Movement to embedded Spec,TP indicates that the subject has the features enabling it to interact with T. We thus cannot explain the absence of interaction with matrix T (hyperraising) via relativized probing. Something else must prevent further movement. To explain the optionality of hyperraising in Zulu, Zeller posits that subjunctive CPs are optionally transparent for A-movement. A transparent CP will give rise to hyperraising, as in (62). If a non-transparent C is merged, we derive (67). On the assumption that hyperraising involves movement through Spec,CP, an opaque CP could simply be implemented as headed by a C without EPP. The derivation for (67) is given in (68). The subject does have [MF] because it moves to embedded Spec,TP. The lack of interaction with matrix T is then not due to the lack of a matching goal but because of a phasal boundary between the matrix T and the embedded subject.

(68)



The other intermediate position available for the subject, shown in (69), posits a much bigger challenge for the relativized probing account. Here, the embedded subject has successfully vacated the embedded CP and landed in the matrix clause. Nonetheless, the subject does not interact with the matrix T – it neither controls agreement on T, nor does it move to Spec,TP.

- (69) Ku/*ba-jayeLe **abafana**_i [_{CP} ukuthi t_i ba-pende t_i indlu.]
15/*2-usually 2boys COMP 2-paint.SBJV 9house
'The boys usually paint the house.'

It is difficult to pinpoint the exact position of the DP 'boys' in (69), but it does appear to be a position inside the matrix clause, rather than the embedded Spec,CP position. This is due to the possibility of being linearized to the left of a matrix adverb (*kakhulu* 'very'):

- (70) Ku-jayeLe **uZodwa** kakhulu [_{CP} ukuthi a-pheke phandle.]
15-usually 1Zodwa very COMP 1s-cook.SBJV outside
'It's very common for Zodwa to cook outside'.

Further, it is quite clear that it is a vP-internal position. First, only vP-internal subjects fail to control agreement on T. Right dislocated subjects also follow the verb, but they obligatorily control agreement on T (71).

- (71) a. Ku/*u-Ø-pheka [_{VP} uZodwa inyama].
 15s/*1s-CNJ-cook 1Zodwa 9meat
 ‘Zodwa cooks meat.’
 b. U/*ku-Ø-pheka [_{VP} t_i inyama] uZodwa_i.
 1s/*15s-CNJ-cook 9meat 1Zodwa
 ‘Zodwa cooks meat.’

Second, the subject position in (69) is compatible with focused expressions such as *wh*-phrases (72-a). These phrases are impossible in dislocated positions (72-b).

- (72) a. Ku-jayeke bani_i [_{CP} ukuthi t_i a-pende t_i indlu?]
 15s-usually 1who COMP 1s-paint.SBJV 9house
 ‘Who usually paints the house?’
 b. *U-za-pheka inyama (u)-bani?
 1s-FUT-cook 9meat (A)-1who
 ‘Who will cook meat?’

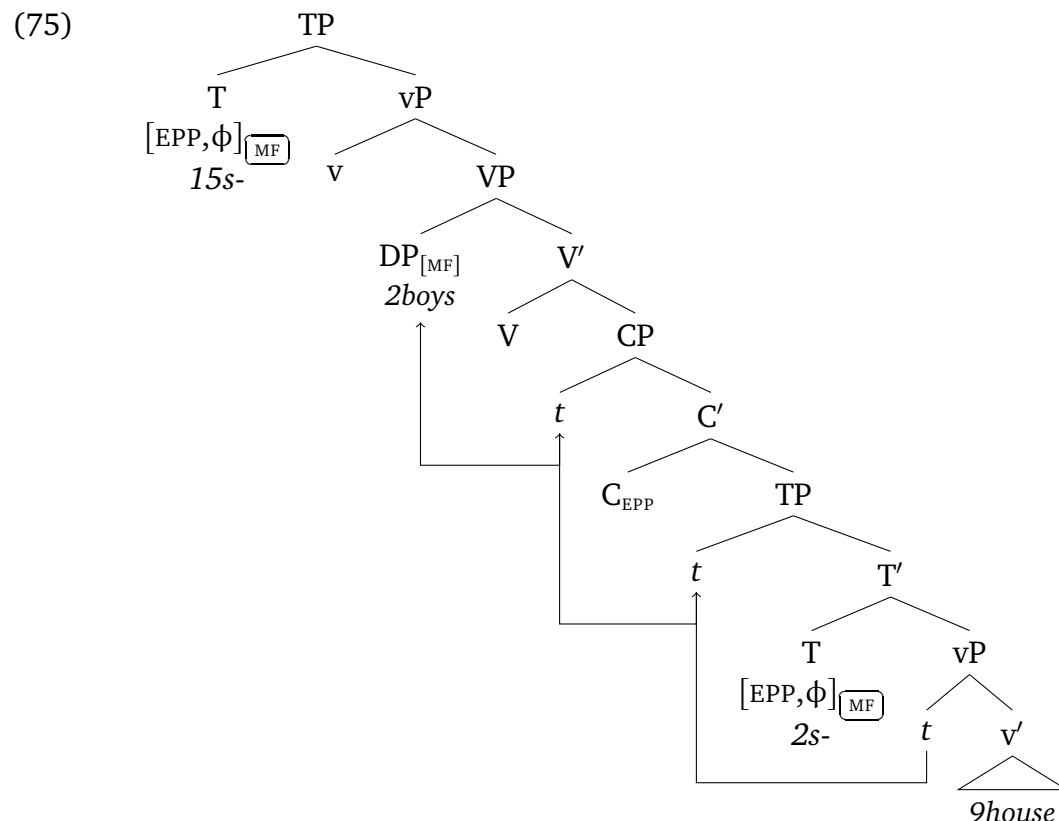
Finally, note that the *wh*-phrase in (72-a) lacks the augment vowel (*bani* instead of *u-bani*). Augment drop has been identified as a diagnostic for a vP-internal position in Ndebele and other Nguni languages (Halpert 2012, 2015; Carstens & Mletshe 2016; Pietraszko 2021). All these properties – lack of agreement, possible narrow focus and augment drop – are characteristic of in-situ subjects. Nonetheless, the subject in (69) is in a derived position, as supported by the truth conditional equivalence of the active and passive counterparts in (73) and the preservation of idiomatic meaning in (74).

- (73) a. Ku-jayeke **abafana**_i [_{CP} ukuthi t_i ba-pende t_i indlu.]
 15s-usually 2boys COMP 2s-paint.SBJV 9house
 ‘The boys usually paint the house.’
 ≈
 b. Ku-jayeke **indlu**_i [_{CP} ukuthi t_i i-pendwe t_i ng-abafana.]
 15s-usually 9house COMP 9s-paint.PSV.SBJV by-2boys
 ‘The house is usually painted by the boys.’
 (74) Ku-jayeke **isigogo**_i [_{CP} ukuthi t_i si-goqwe t_i sisemanzi.]
 15-usually 7leather COMP 7-fold.PSV.SBJV wet.PTCP
 Lit: ‘Leather is usually folded while still wet’
 Idiom: ‘Something/it is usually done at the right time.’ ✓

We conclude that hyperraising to subject can target a vP-internal position in Ndebele.¹⁸

¹⁸A reviewer asks what triggers hyperraising to subject to terminate inside vP. On the approach adopted here, A-movement is triggered solely by the presence of an EPP feature. No additional motivation, such as the need to license case on the DP, is required for movement to occur. While case often correlates with movement in other languages, the link between the two is much less clear in Bantu. In fact, the existence of case in Bantu languages have been questioned (e.g., Harford Perez 1985; Diercks 2012; Carstens & Diercks 2013) or argued to be substantially different from other case systems (Halpert 2015; Sheehan & van der Wal 2018; Pietraszko 2021). Under the current analysis, the relevant A-movement feature (EPP) may be merged in different positions (e.g., V or T), and it is this feature alone that drives the

This is perhaps not surprising, given that the language has hyperraising to object (see e.g. (46)). I will tentatively assume that the vP-internal position targeted by raising is always Spec,VP, whether the raised DP ends up the matrix subject or object.¹⁹ Nothing hinges on this specific location however, as long as the position is within the vP. With these pieces in place, the derivation of (69) under the relativized probing account is given in (75).



The derivation in (75) looks incomplete: The matrix T is searching for a DP with [MF] and such a DP is present and accessible – the DP *boys* raised from the embedded clause to matrix VP. This DP has successfully interacted with the embedded T, so the lack interaction with matrix T cannot be explained by the lack of a matching goal for T. Similarly, locality should not be an issue here. A clause-internal phase plays no role in regulating subject movement on Zeller’s account, and the CP has already been crossed.

In sum, the relativized probing account of the SV-VS alternation incorrectly predicts

movement.

The reviewer also asks whether hyperraising to Spec,TP must proceed via the matrix vP-internal position. I am not aware of a reliable diagnostic that could confirm or rule out the presence of this intermediate step. If raising to Spec,TP were to bypass the lower A-position, it would require an account of why EPP is absent from V in these derivations. Barring such an explanation, the more parsimonious view is that matrix V in raising constructions always bears EPP, irrespective of whether the movement terminates in Spec,VP or continues on to Spec,TP.

¹⁹ Assuming that *subject* is defined as the highest DP in its clause, raising to object and raising to a vP-internal subject position may target the same structural position (e.g., Spec,VP). Whether the raised DP surfaces as the matrix subject or object then depends on whether it ends up as the highest DP in the matrix clause.

The phasal VoiceP account allows raising to VP with no further modifications. Movement to Spec,TP is regulated in the same way in the matrix and the embedded clause: whether it takes place depends on whether the local Voice has EPP. Raising to VP occurs when the embedded Voice has EPP but the matrix Voice does not (76).

35

- (77) a. No movement
 $[T_{[\phi, EPP]} [Voice_{(EPP)} \dots [C_{(EPP)} [T_{[\phi, EPP]} [Voice [Zodwa\ v]]]]]]]$
- b. Movement to embedded Spec,TP
 $[T_{[\phi, EPP]} [Voice_{(EPP)} \dots [C [Zodwa_i T_{[\phi, EPP]} [t_i Voice_{EPP} [t_i\ v]]]]]]]$
- c. Movement to matrix Spec,TP
 $[Zodwa_i T_{[\phi, EPP]} [t_i Voice_{EPP} \dots [t_i C_{EPP} [t_i T_{[\phi, EPP]} [t_i Voice_{EPP} [t_i\ v]]]]]]]$

(77-a) shows the derivation of sentences such as (65), where the subject remains in the base-position. This occurs when the embedded Voice lacks EPP. The features on C and matrix Voice are irrelevant in this case – even if these heads have EPP, they would not be able to access the in-situ subject. Movement terminating in embedded Spec,TP, as in (67), is derived when embedded Voice has EPP but C does not. Finally, movement all the way to matrix Spec,TP (62) requires EPP on all intervening phase heads: embedded Voice, C, and matrix Voice.

The paradigm in (77), combined with our observations from Aux-V constructions, allows us to make an interesting generalization about where A-movement is optional and where it is obligatory. Our findings in the respect are summarized in (78).

- (78) $[vP \xrightarrow{\text{optional}} [CP_{\text{phase}} \xrightarrow{\text{oblig.}} TP \xrightarrow{\text{oblig.}} PerfP \xrightarrow{\text{optional}} [VoiceP_{\text{phase}} \xrightarrow{\text{optional}} vP]]]]]$

We saw in section 3.1.3 that movement from the base position inside vP is optional, but once the subject vacates vP and lands in AspP, for instance, movement obligatorily proceeds to PerfP and to TP (if present). Another optional movement is across a CP, as in hyperraising. Obligatoriness of movement is then delimited by phase boundaries: movement is obligatory within a phase and optional across a phasal category. This generalization is stated in (79).

- (79) *Phase-delimited movement*

In Ndebele, obligatoriness of A-movement is predictable from phase boundaries:

- i. Phase-internal movement is obligatory
- ii. Cross-phasal movement is optional

This generalization provides further support for the claim that there are two opacity domains in the Ndebele clause as it allows us to capture the domains of obligatory movement in the language.

In conclusion, this section has argued against the relativized probing analysis of the SV-VS alternation, according to which the subject is not always a suitable goal for T. As revealed by hyperraising to VP, the subject can fail to move to Spec,TP even when it demonstrably has all the features that T is looking for. Some other mechanism must therefore account for the optional absence of subject movement to Spec,TP. A clause internal phase provides such a mechanism.

5 The present findings in broader perspective

The conclusion that VoiceP is phasal in Ndebele aligns with a substantial body of work supporting the existence of a clause-internal phase. Nonetheless, certain questions remain open. In this section, I address three issues that arise in connection to ongoing debates about the nature of phases. The first concerns the category of the clause-internal phase: while the analysis here assumes it is VoiceP, earlier proposals have typically identified it with vP (5.1). The second relates to a growing body of literature that questions the existence of any clause-internal phase, citing apparent transparency to movement or agreement as evidence (5.2). Finally, I offer a brief remark on the implications of the present findings for theories of successive cyclic movement, particularly its triggers and obligatoriness (5.3).

5.1 The category of the clause internal phase

The present analysis crucially assumes that the relevant phasal projection is located above the position that introduces the external argument, a claim that appears to conflict with the traditional view that vP, typically the projection that introduces the external argument, is the clause-internal phase. The concern is that this departs not only from early Minimalist assumptions, but also from specific analyses where the external argument itself occupies the edge of the phase, explaining its unique accessibility to certain syntactic operations. Although I cannot offer a unified analysis of all cases in which vP appears to be phasal, it is important to recognize that the category of the clause-internal phase has undergone substantial revision in recent theoretical work.

The original proposal that vP is a phase ([Chomsky 2000, 2001](#)) was based on a simplified clause structure with no projections like VoiceP or ProgP intervening between vP and TP. However, subsequent research has argued that the clause-internal phase is higher than vP, and corresponds instead to VoiceP or ProgP (i.a. [Aelbrecht & Harwood 2015](#); [Harwood 2015a](#); [Ramchand 2018](#)). Building on this body of work, [Sheehan & Cyrino 2024](#) argue that in both English and Brazilian Portuguese, the external argument is contained within the phasal complement and is crucially not located at the phase edge. The same conclusion was reached by [Pietraszko \(2023b\)](#) for Ndebele, based on the uniform behavior of agents and other types of arguments.

The broader crosslinguistic variation in where the clause-internal phase appears to be located (vP, VoiceP, ProgP, etc.) may reflect genuine differences across languages, or may arise from how phasehood is defined. One possibility is that phasehood is determined not by fixed syntactic labels but by derivational factors such as agreement or feature valuation, leading to context-sensitive or dynamic phasehood ([Rackowski & Richards 2005](#); [Abels 2003](#); [Bobaljik & Wurmbrand 2005](#); [Gallego 2007](#); [Gallego & Uriagereka 2008](#); [Takahashi 2011](#); [Bošković 2014](#); [Wurmbrand 2017](#), among others). Another possibility is that the privileged status of the external argument, its apparent ability to move or agree across boundaries that are opaque for lower arguments, does not follow from its position at the phase edge, but from independent principles, such as relativized minimality. That is, at least some cases in which the external argument exhibits special accessibility may be attributed to its status as the highest DP in the thematic domain, and need not indicate

whether it is generated at the phase edge or within the phasal complement.

Determining which of these possibilities applies in a given language or construction is ultimately an empirical matter. The analysis I advocate here is in principle consistent with either view. VoiceP in Ndebele exhibits the core empirical properties associated with phasehood, regardless of whether that phasal status is fixed or contextually derived.

5.2 Reconciling the opacity of VoiceP with evidence for transparency elsewhere

The conclusion that VoiceP is a phase in Ndebele raises important questions about how to reconcile this with findings from other languages, where comparable projections (typically vP) appear transparent to movement or agreement. Several recent studies report that vP does not exhibit the expected hallmarks of a phase, such as footprints of successive cyclic movement or locality-based agreement restrictions, and thus conclude that it lacks phasal status. However, these conclusions often arise not from direct evidence against vP phasehood, but from the need to assume a non-phasal vP in order for a particular analysis to go through. Moreover, these accounts tend not to explore alternative models in which vP is a phase but rendered transparent through other mechanisms.

A number of recent works, such as [Grano & Lasnik 2018](#); [Keine 2017, 2020a](#), [Mendes & Ranero 2021](#), conclude that vP is not a phase, either because a phasal vP would incorrectly block a dependency (as in [Grano & Lasnik 2018](#)), or because there is no evidence for the kind of intermediate movement or locality effects that would be expected if vP were phasal (as in the other three). [Grano & Lasnik](#) argue that cross-clausal pronominal binding renders the intervening CP transparent. Their explanation for this transparency is incompatible with phasal vP, which would introduce additional opacity that doesn't follow from their account. [Mendes & Ranero](#) observe that the particle *wi* in Mayan appears only at CP edges, not vP edges, which they take to indicate that movement does not proceed through vP. [Keine \(2020a\)](#), drawing on processing evidence, argues that intermediate reactivation effects – taken as signatures of successive cyclic movement – are found at CP edges but not at vP edges. And [Keine \(2017\)](#) presents evidence from Hindi that both ϕ -agreement and *wh*-licensing can cross multiple vP domains, in apparent violation of the Phase Impenetrability Condition. Taken together, these studies suggest that vP does not behave as a phase, either because dependencies are not blocked by it, or because intermediate movement through its edge is not observed.

A shared limitation of these accounts is that they do not consider alternative analyses of the phenomena in question that would be compatible with a phasal vP. The conclusion that vP is not a phase is introduced as a requirement of the proposed analysis, rather than as the outcome of a broader theoretical comparison. For example, [Grano & Lasnik](#) state that their proposal is “a better fit with the view that CPs are phases (under some conditions) whereas vPs are not” (p. 482), but they do not explore alternative explanations for the transparency of CPs containing bound pronouns, and whether such alternatives could be reconciled with a phasal vP. As a result, the evidence these works offer against vP phasehood remains weak: it shows only that assuming vP to be phasal is incompatible with the success of a specific analysis. To be clear, a well-formulated analysis that succeeds only under the assumption that some hypothesis X is false (here, phasal vP) may offer indirect evidence against X. But for this to constitute genuine evidence, it must be shown that no plausible alternative

analysis, one compatible with X, can account for the same data with similar success.

Keine (2017) presents a stronger empirical challenge to vP phasehood, based on the observation that ϕ -agreement and *wh*-licensing in Hindi can span multiple vP domains without triggering the locality effects typically associated with phases. Since agreement is standardly modeled as a probe-goal relation subject to locality constraints, the apparent transparency of vP poses a more serious problem for its phasal status. However, even in this case, the analysis does not consider whether the relevant dependencies could be accounted for within a phase-based framework – for example, one in which vP is unlocked for agreement. That is, the absence of phasehood effects is treated as evidence that vP is not a phase, but not weighed against phase-compatible mechanisms that could produce the same empirical outcome – though in the case of agreement, the range of alternatives may be narrower, given the existence of a relative consensus about how agreement works.

What none of these studies consider is that movement or agreement might bypass the phase edge altogether – not because the domain is not phasal, but because it has been rendered transparent under specific syntactic conditions. This possibility is central to more recent developments in phase theory, which treat phasehood not as a fixed property of specific projections, but as a probe-relative, context-sensitive phenomenon that can vary across constructions and languages. A prominent line of work has argued that phases can be rendered transparent through a mechanism known as phase unlocking (Rackowski & Richards 2005; Halpert 2019; Branen 2018; Ershova 2024). On this view, a phase becomes accessible to a higher probe when it has entered into an Agree relation with that probe. Once such agreement occurs, the phase ceases to intervene, allowing movement or further agreement into its complement without requiring movement through the edge. This mechanism provides a way to reconcile successful extraction across a clause-internal phase without abandoning its phasal status or requiring movement through its edge. From this perspective, the absence of successive cyclic movement through vP – whether inferred from syntactic patterns or processing data – does not entail that vP is not a phase. The transparency of vP observed in some languages/constructions, can be compatible with the opacity of VoiceP in Ndebele, provided that we adopt a more flexible, dynamic view of phasehood.²⁰

Although Pietraszko (2023b)’s analysis does not adopt a phase-unlocking framework to derive the subject movement and agreement, it’s possible it could be reframed in those terms: VoiceP would remain a phase, but its transparency could depend on the presence of some feature – functionally parallel to the optional EPP in Voice. If such a feature is present, VoiceP becomes transparent; if not, it acts as a boundary. This reframing would preserve

²⁰One might worry that if a phase-neutralizing mechanism, such as phase-unlocking, were always available, then any XP could be claimed to be a phase, rendering phasehood unfalsifiable. But this isn’t how a well-formulated phase theory would work. An analogy to CPs in current phase theory is useful here. CP opacity is neutralized in English *wh*-movement by a successive cyclic step, yet the phasehood of CP is revealed, for example, in A-movement, where opacity surfaces directly. In other words, A-movement shows that the transparency of CP for A-bar movement is due to the neutralization of opacity, not due to CP not being a phase. The same logic should apply to any other neutralizing mechanisms we might want to introduce to the theory: they must be constrained, applying only in certain configurations, with opacity visible elsewhere. This is not only a theoretical necessity to avoid unfalsifiability, but also an empirical one: the fact that we observe operational opacity in many languages (e.g. CP for A-movement in English, VoiceP for ϕ -agreement and A-movement in Ndebele) shows that such mechanisms cannot be freely available.

the broader conclusion defended in this paper: VoiceP is a phase, and its transparency must be independently derived, whether through successive cyclic movement or through agreement-driven unlocking.

In sum, while the transparency of vP in some languages has been taken as evidence against its phasal status, this conclusion is not inescapable. As discussed above, such transparency can arise from a range of factors, including the presence of unlocking mechanisms or variation in which projection constitutes the clause-internal phase. Sorting out which explanation applies in which context remains an open question, and one that lies beyond the scope of this paper. My goal here has been to highlight that multiple analytical paths remain available, and that apparent transparency effects are not in themselves sufficient to rule out the existence of a clause-internal phase.

5.3 Implications for successive cyclic movement

The present results corroborate the conclusion reached in [Sheehan & Cyrino 2023](#) that A-movement cannot freely cross phasal boundaries. A common assumption about successive cyclic movement is that it is always available whenever a phase boundary separates a probe and a matching goal (modulo other issues such as *Doubly-Filled Comp). This has been implemented in two ways: i) phase heads are inherently equipped with a movement feature (e.g. Edge Feature, [Chomsky 2000](#)) or ii) movement through phase edge is free to apply without a featural trigger (i.a. [Bošković 2002, 2007](#); [Chomsky 2013](#)). Either implementation would allow A-movement in Ndebele to freely cross phasal VoiceP, predicting that the subject should never stay in situ. Instead, movement through Spec,VoiceP in Ndebele is like any other movement: it takes place iff a movement probe is merged. For this reason, the analysis proposed here uses the same feature (EPP) for both intermediate movement and movement to the final landing site. However, this conclusion is not incompatible with the existence of a special feature that drives exclusively intermediate movement ([Georgi 2014](#)). The claim here is weaker: Whatever feature licenses movement through a phase edge, it is not a universal property of phase heads.

6 Conclusion

This paper defended the clause-internal phase hypothesis by arguing that VoiceP in Ndebele is opaque for A-movement and ϕ -agreement. Alternative accounts that do not invoke a clause internal phase were ruled out on empirical grounds. This conclusion adds to a body of existing literature that supports the clause internal phase, but is crucially immune to a lot of the criticism that this literature came under. This is because the evidence presented here does come from footprints of intermediate movement, which are logically compatible with the absence of intervening phases ([Adger & Ramchand 2005](#), [den Dikken 2009, 2017](#), [Keine & Zeijlstra](#) to appear). In Ndebele, we observe the phasehood of VoiceP more directly, by its opacity to syntactic operations.

Many questions remain open, as the debate surrounding the clause-internal phase is marked by conflicting evidence – both regarding the precise locus of the phase and its existence in the first place. The conclusion that VoiceP functions as a phase in Ndebele should not be taken to imply that clause-internal phase effects must manifest uniformly

across languages. My aim has been to show that the Ndebele data strongly support a phase-based analysis, based on empirical contrasts with competing accounts that do not posit a clause-internal phase. Evidence of this kind suggests that observed transparency in other languages need not indicate the universal absence of a clause-internal phase, and is more naturally understood within a theoretical framework that allows for conditional transparency.

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